UTS

DVN36104 SPR 2026

Data Visualisation and Narratives
Spring 2026
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UTS CRICOS 00099F
UTS TEQSA PRV12060

Acknowledgment of Country

Image Source: https://pixabay.com/photos/aboriginal-art-aboriginal-painting-503445/

What to expect in this Subject?

- Justify the selection and analysis of data as the basis for ‘data narratives’ for different stakeholders.
- Apply a range of visualisation and narrative techniques to a variety of data types.
- Justify the selection of narrative tools and techniques to illuminate critical aspects of problems.
- Justify and communicate ‘data narratives’ to stakeholders from a range of industries and contexts drawing on relevant data patterns and analyses.

I can’t → I can’t yet / maybe I can → I can confidently

In practical terms, you’ll learn to …

- be selective to qualm, appease stakeholders
- strengthen your mastery of tools
- expose, highlight, amplify, influence.
- understand patterns, adopt, adapt, model, scale

I can’t → I can’t yet / maybe I can → I can confidently

DVN is all about You

- You acting to learn about opportunities with data, visuals, and making something visual
- It’s about you being open and expressive when telling a story
- Education is a team effort, so feedback is key for us
- Participate
    - Ask questions
    - Contribute ideas and your perspectives
    - Complete in-class activity, tutorials and exercises
- Assignments are to reinforce learnings

Let’s break it down

DATA VISUALISATION

- "data" (the information itself)
- "visual" (presented in a visual format)
- "isation" (the act of making something visual)

NARRATIVES

- Narr - This is the root of the word, meaning "to tell a story."
- a - This is a suffix that often indicates an action or process.
- tive - This suffix means "relating to" or "having the quality of."

Learning Objectives – Week 1

- Gain knowledge of data types, formats, and big data frameworks
- Understand the importance of context in data visualisation
- Differentiate between explanatory and exploratory data visualisation
- Select effective visuals for data representation
- Identify types of visuals according to Edward Tufte's Framework
- Define and extract meaningful insights from data

Image Generated by Co-pilot

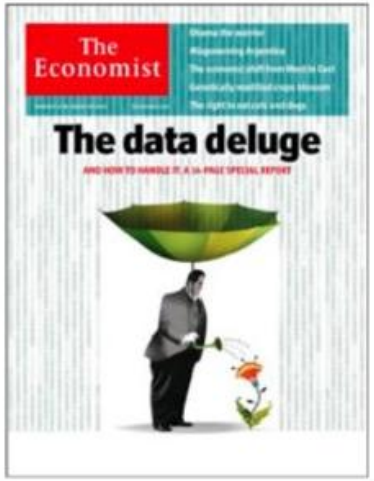
Section 1

Understanding Data for Visualisation

Data

Data is information that has been translated into a form that is efficient for movement or processing

Data Everywhere!



"The ability to take data - to be able to understand it, to process it, to extract value from it, to visualise it, to communicate it's going to be a hugely important skill in the next decades. Because now we really do have essentially free and ubiquitous data.

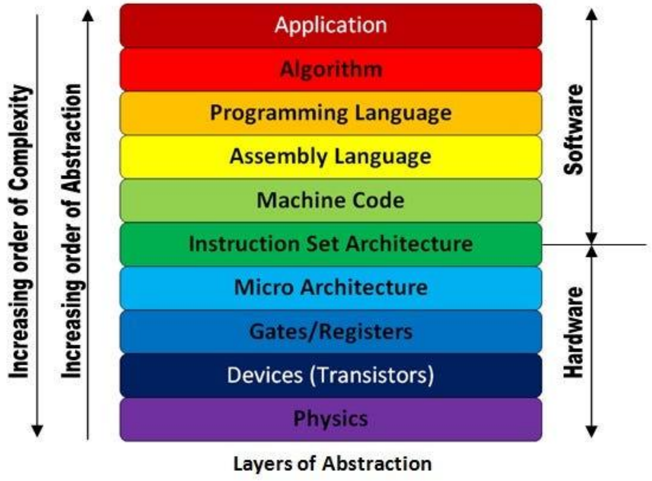
So the complimentary scarce factor is the ability to ***understand that data and extract value from it.****"*

Hal Varian, Chief Economist of Google

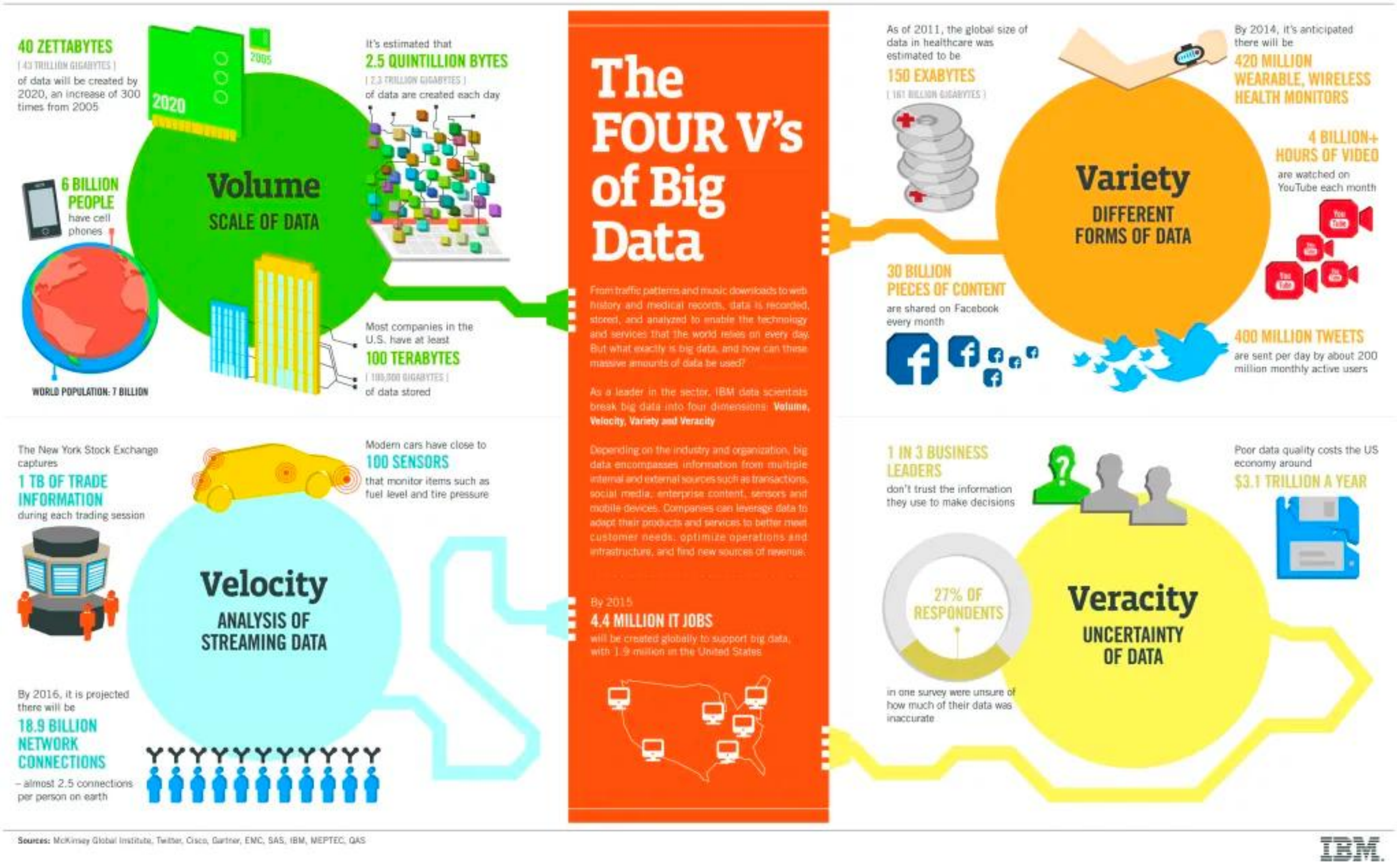
Source: https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/hal-varian-on-how-the-web-challenges-managers

Data is everywhere

- Text, images, audio, videos data are 0s and 1s but with **abstraction** applied
- Collected, parse, transformed and stored on application layer in widely connected, smart devices or servers in data centres
- Didn’t and won’t magically evolve



4 Vs' of big data (2015)



Source: https://www.ibm.com/think/topics/big-data

Understanding big data and its associated challenges

Volume: Massive data sizes (terabytes, petabytes).

Velocity: High-speed data generation.

Variety: Different data types and formats.

Veracity: Reliability and accuracy of data.

A fifth V

Oracle proposed in 2024 that:

- Data has intrinsic value in business, but its value must be discovered.
- Big data provides extensive insights that can benefit an organization.
- Insights can optimise internal processes or enhance external customer engagement.

VALUE

Source: https://www.oracle.com/au/big-data/what-is-big-data/

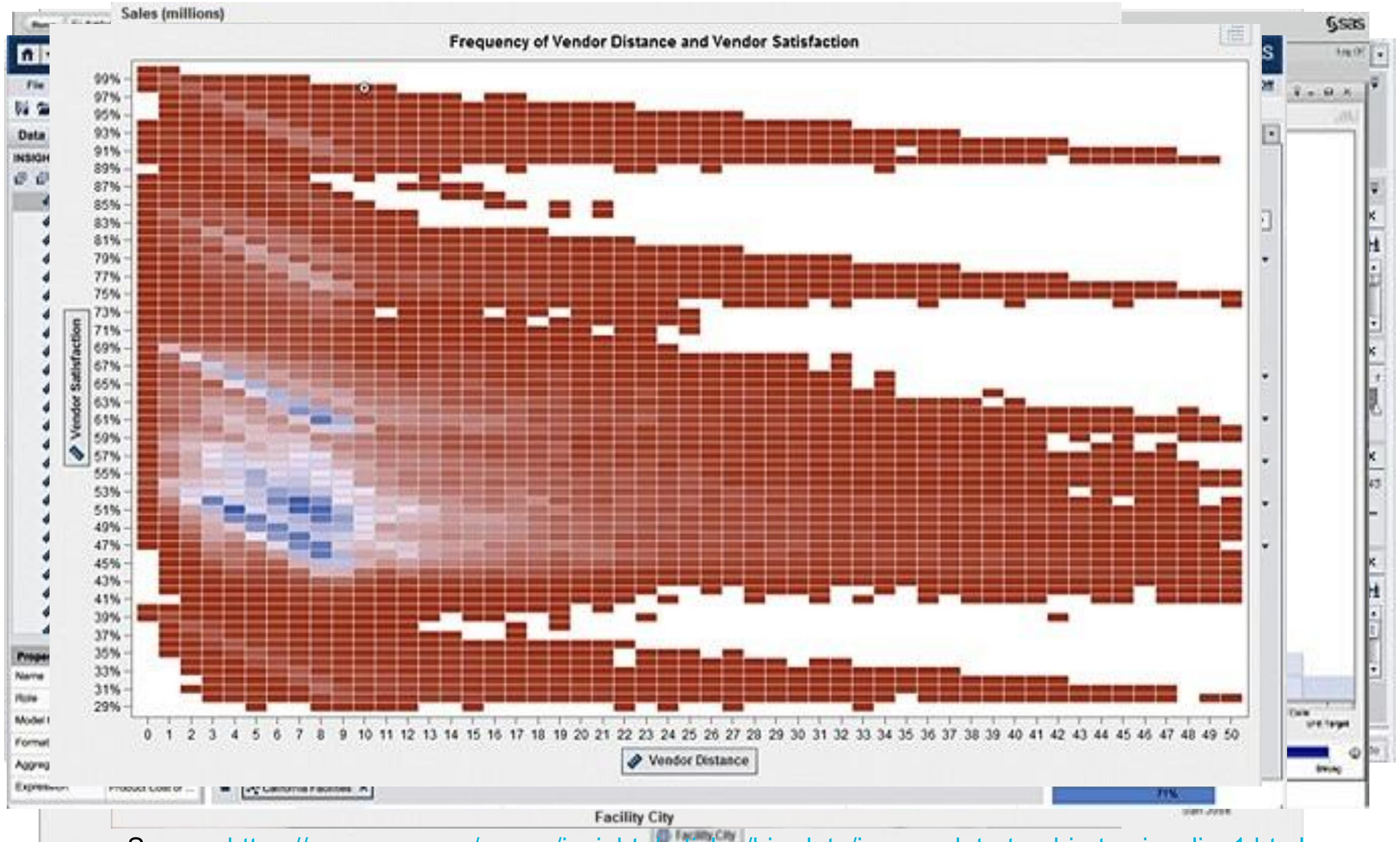
What Is visual encoding?

Position	Length	Colour	Size	Shape	Orientation	Texture
●	▮	■■■	• ● ⬤	▲	///	
where something sits on an axis	bars, lines, distances	hue, intensity, gradients	area, radius, thickness	categories, markers	direction, angle	density, variation

Source: https://www.sas.com/en_au/insights/articles/big-data/is-your-data-too-big-to-visualize1.html

Six abstraction techniques in big data visualisation

1. Filtering
2. Binning
3. Autocharting (SAS Visual Analytics, Tableau Prep, Alteryx)
4. Network Diagrams
5. Overview Bars for high Cardinality (50 categories)
6. Correlation Matrices

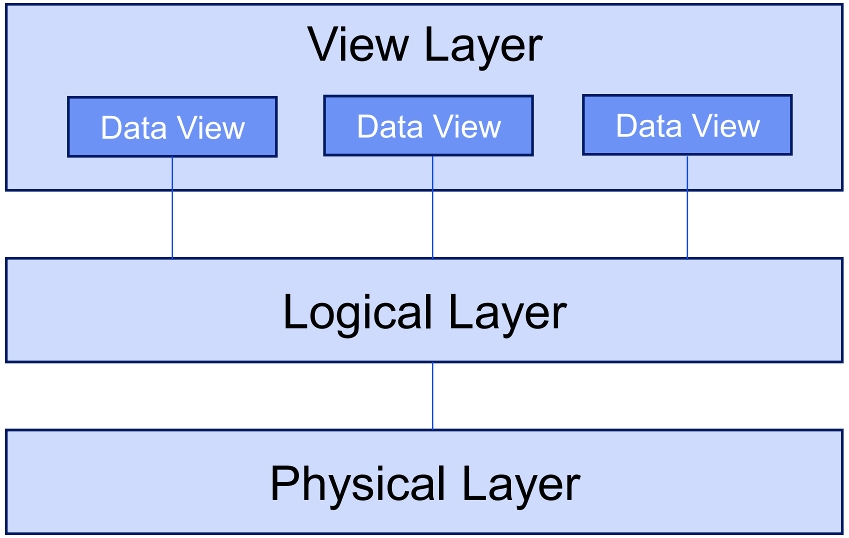


Source: https://www.sas.com/en_au/insights/articles/big-data/is-your-data-too-big-to-visualize1.html

Data abstraction in our data viz context

- **Purpose:**
  - Hides the complexities of data storage and management in a Database Management System (DBMS).
  - Provides different levels of data views for various users.

- **Three Layers of Data Abstraction:**
  - **View Layer:**
    - User-facing layer, showing data as presented by applications.
    - Users typically have limited access and understanding of the complete data.
    - Focuses on *how* data is presented.
  - **Logical Layer:**
    - Conceptual layer, describing data types and relationships.
    - Represents the data structure (e.g., entity-relationship diagrams).
    - Focuses on *what* data is stored.
  - **Physical Layer:**
    - Deals with physical data storage details (location, file management, storage type).
    - Focuses on *where* data is stored.



Data types..

Structured Data: Organized in fixed formats, typically in databases (e.g., SQL databases).

Semi-Structured Data: Contains tags or markers to separate elements but doesn't conform strictly to a schema (e.g., JSON, XML).

Unstructured Data: Lacks a pre-defined structure (e.g., text documents, images).

..in our context

Structured: Spreadsheet, SQL Database

Semi-structured: XML, JSON

Date	Product Name	Customer ID	Quantity	Price	Total	Region	Sales Rep
2024-07-26	Laptop Pro X1	1001	2	1200	2400	North	John Smith
2024-07-26	Office Suite Software	1002	1	300	300	East	Jane Doe
2024-07-27	Wireless Mouse	1001	5	25	125	North	John Smith
2024-07-27	Ergonomic Keyboard	1003	1	150	150	West	David Lee
2024-07-28	Laptop Pro X1	1002	1	1200	1200	East	Jane Doe
2024-07-28	Monitor 27"	1004	2	350	700	South	Sarah Jones
2024-07-29	Office Suite Software	1001	3	300	900	North	John Smith
2024-07-29	Wireless Headphones	1005	1	200	200	West	David Lee

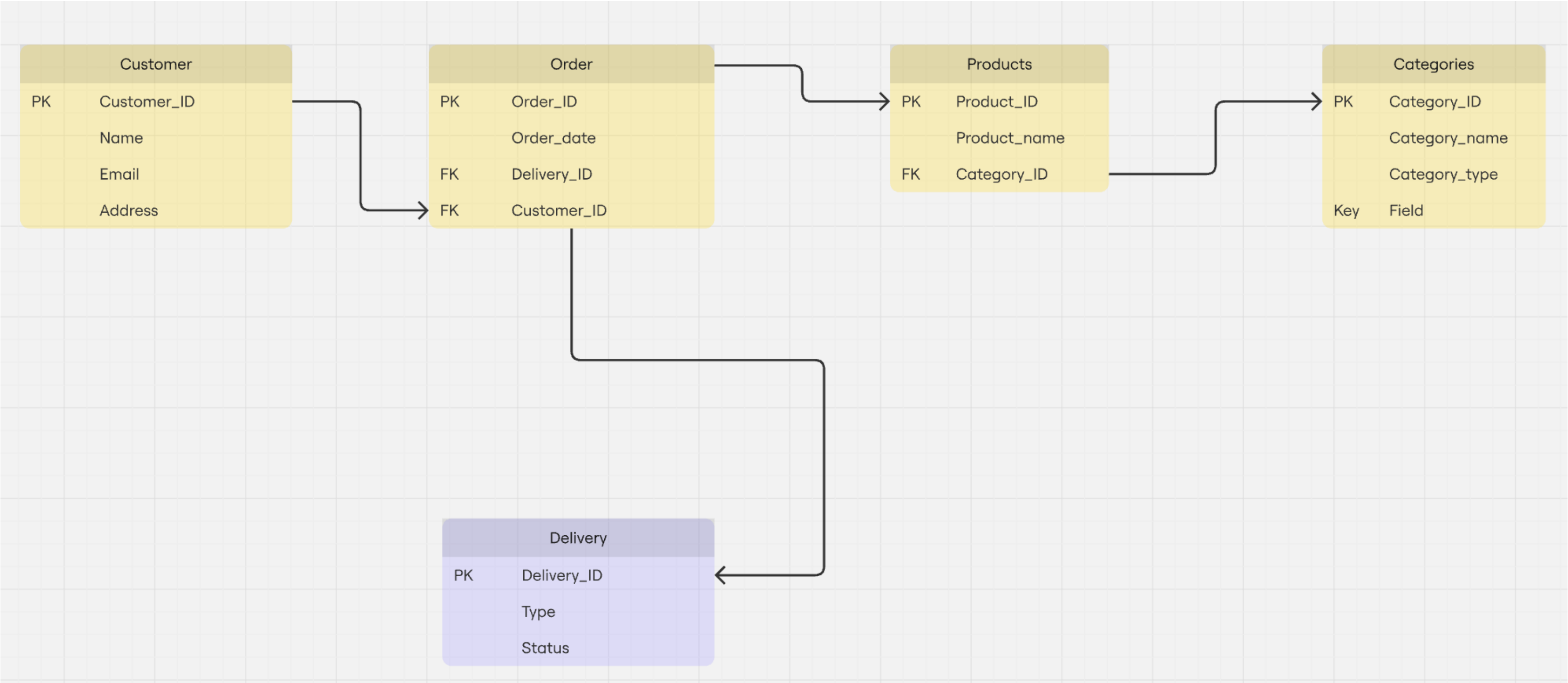

```xml
<?xml version="1.0" encoding="UTF-8"?>
<library>
  <book category="COOKING">
    <title lang="en">Everyday Italian</title>
    <author>Giada De Laurentiis</author>
    <year>2005</year>
    <price>30.00</price>
  </book>
  <book category="CHILDREN">
    <title lang="en">Harry Potter and the Sorcerer's Stone</title>
    <author>J.K. Rowling</author>
    <year>2005</year>
    <price>29.00</price>
  </book>
  <book category="WEB">
    <title lang="en">XQuery Kick Start</title>
    <author>James McGovern</author>
    <year>2003</year>
    <price>49.99</price>
  </book>
  <book category="WEB">
    <title lang="en">Learning XML</title>
    <author>Erik T. Ray</author>
    <year>2003</year>
    <price>39.95</price>
  </book>
</library>
```


```sql
-- Sample Data
INSERT INTO Constellations (ConstellationID, Name, Abbreviation) VALUES
(1, 'Orion', 'Ori'),
(2, 'Ursa Major', 'UMa'),
(3, 'Andromeda', 'And');

INSERT INTO Stars (StarID, Name, ConstellationID, RightAscension, Declination, DistanceLightYears) VALUES
(1, 'Betelgeuse', 1, 5.88, 45.0, 642.5),
(2, 'Polaris', 2, 2.53, 89.3, 433.8),
(3, 'Sirius', 0, 101.3, -16.7, 8.6);  -- Sirius is not in a named constellation

INSERT INTO Planets (PlanetID, StarID, Name, OrbitalPeriodDays, MassRelativeToEarth) VALUES
(1, 1, 'Betelgeuse b', 2000, 5.0),  -- Hypothetical planet
(2, 3, 'Sirius b', 50 * 365, 0.98); -- Hypothetical planet


INSERT INTO Galaxies (GalaxyID, Name, Type, DistanceLightYears) VALUES
(1, 'Andromeda Galaxy', 'Spiral', 2500000),
(2, 'Milky Way', 'Spiral', 0);  -- Our own galaxy, so distance is 0

INSERT INTO Observations (ObservationID, StarID, GalaxyID, ObservationDate, Observer, Notes) VALUES
(1, 1, NULL, '2024-08-01', 'Dr. Smith', 'Observed a slight increase in brightness'),
(2, NULL, 1, '2024-08-05', 'Jane Doe', 'Detailed images captured');
```


```json
[
  {
    "sensorId": "S001",
    "timestamp": "2024-10-27T10:00:00Z",
    "temperature": 25.5,
    "humidity": 60.2,
    "location": {
      "latitude": 37.7749,
      "longitude": -122.4194
    },
    "status": "active"
  },
  {
    "sensorId": "S002",
    "timestamp": "2024-10-27T10:00:15Z",
    "pressure": 1012.3,
    "altitude": 150.7,
    "location": {
      "latitude": 37.7750,
      "longitude": -122.4195
    },
    "status": "active"
  },
  {
    "sensorId": "S001",
    "timestamp": "2024-10-27T10:00:30Z",
    "temperature": 25.6,
    "humidity": 60.5,
    "location": {
      "latitude": 37.7749,
      "longitude": -122.4194
    },
    "status": "active"
  },
   {
    "sensorId": "S003",
    "timestamp": "2024-10-27T10:00:45Z",
    "carbonMonoxide": 2.1,
    "location": {
      "latitude": 37.7751,
      "longitude": -122.4196
    },
    "status": "inactive"
  }
]
```

Relational DB and Entity-Relationship Diagrams (ERD)

- To understand relational databases
  - Tables
    - IDs
    - Field Name
    - Data Type
  - Relationships



JSON/XML and it's fluid schema

- Acts as a data dictionary
  - Field Name
  - Field Description
  - Field Type

JSON

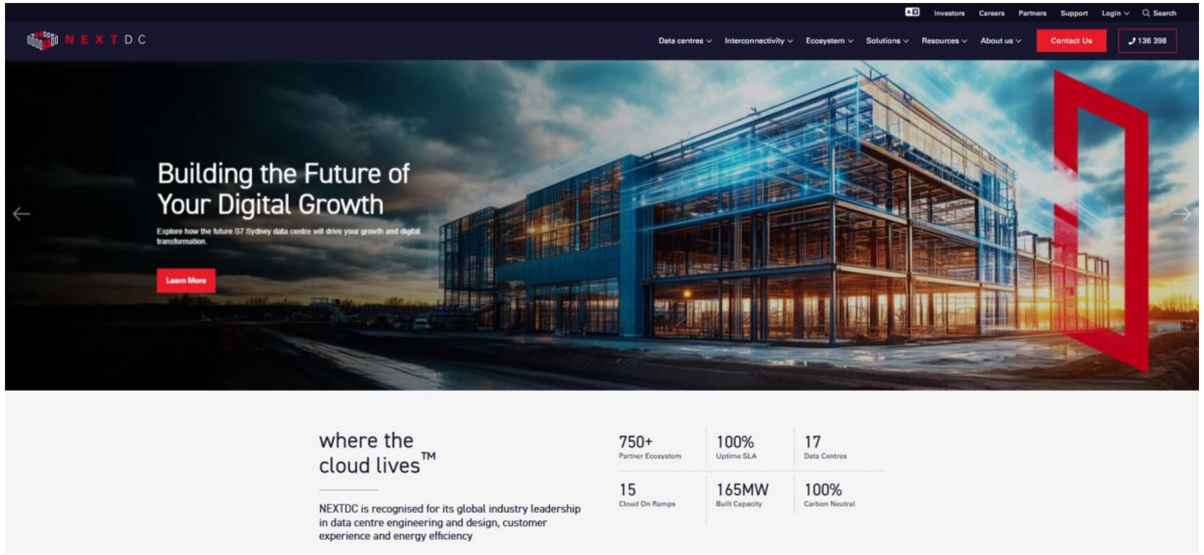

```json
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "Sales Record",
  "description": "Schema for sales transactions",
  "type": "object",
  "properties": {
    "Date": {
      "description": "Date of the sale",
      "type": "string",
      "format": "date"
    },
    "Product Name": {
      "description": "Name of the product sold",
      "type": "string"
    },
    "Customer ID": {
      "description": "ID of the customer",
      "type": "integer"
    },
    "Quantity": {
      "description": "Quantity of products sold",
      "type": "integer"
    },
    "Price": {
      "description": "Price per unit",
      "type": "number"
    },
    "Total": {
      "description": "Total sale amount",
      "type": "number"
    },
    "Region": {
      "description": "Region of the sale",
      "type": "string"
    },
    "Sales Rep": {
      "description": "Sales representative",
      "type": "string"
    }
  },
  "required": [
    "Date",
    "Product Name",
    "Customer ID",
    "Quantity",
    "Price",
    "Total",
    "Region",
    "Sales Rep"
  ]
}
```


JSON


```json
{
  "Date": "2024-07-26",
  "Product Name": "Laptop Pro X1",
  "Customer ID": 1001,
  "Quantity": 2,
  "Price": 1200,
  "Total": 2400,
  "Region": "North",
  "Sales Rep": "John Smith"
}
```

Unstructured data

- Websites, Documents, Transcripts, Images, Videos
  - Needs to be processed
  - Always validate its authenticity
  - Be a gracious host and attribute source



Data file formats and their use cases

Text-Based Formats: CSV, JSON, XML for transferring data.

Multimedia Formats: JPEG, MP4, WAV for image, video, and audio.

Binary Formats: Optimised for performance and storage efficiency (e.g., Parquet, Avro).

No restrictions in this course but level of complexity increases as you move from left to right

What is data visualisation?

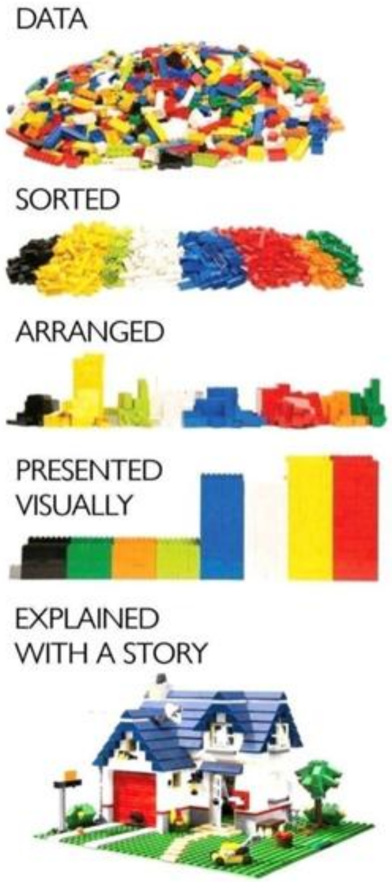
(interactive) **visual** representation of information to help people **make sense** of complex phenomena **through data**.

Data visualisation improves one's

Understanding:
Makes large data sets more accessible, understandable, and usable.

Decision-Making:
Facilitates faster and more informed decisions.

Pattern Recognition:
Allows spotting trends, correlations, and outliers quickly



Activity Time: Favourite Infographic Source (e.g. Voronoi App)

Image Generated by Co-pilot

Section 2

Importance of Context when Narrating Data using Visualisations

Let us look at a historic medical example

Ignaz Semmelweis, a mid-nineteenth century obstetrician, who **discovered hand washing could save countless lives** but **failed to communicate** his findings effectively to a sceptical medical community. His data was ignored, his life-saving ideas were rejected, and he was sadly discredited by his colleagues.

Source: https://www.forbes.com/sites/brentdykes/2016/02/09/a-history-lesson-on-the-dangers-of-letting-data-speak-for-itself/?sh=57d2e24820e1

Data table presented

Table 1. Annual births, deaths, and mortality rates for all patients at the two clinics of the Vienna maternity hospital from 1841 to 1846.

	First Clinic			Second Clinic		
	Births	Deaths	Rate	Births	Deaths	Rate
1841	3036	237	7.7	2442	86	3.5
1842	3287	518	15.8	2659	202	7.5
1843	3060	274	8.9	2739	164	5.9
1844	3157	260	8.2	2956	68	2.3
1845	3492	241	6.8	3241	66	2.03
1846	4010	459	11.4	3754	105	2.7
Total	20 042	1989		17 791	691	
Avg.			9.92			3.38

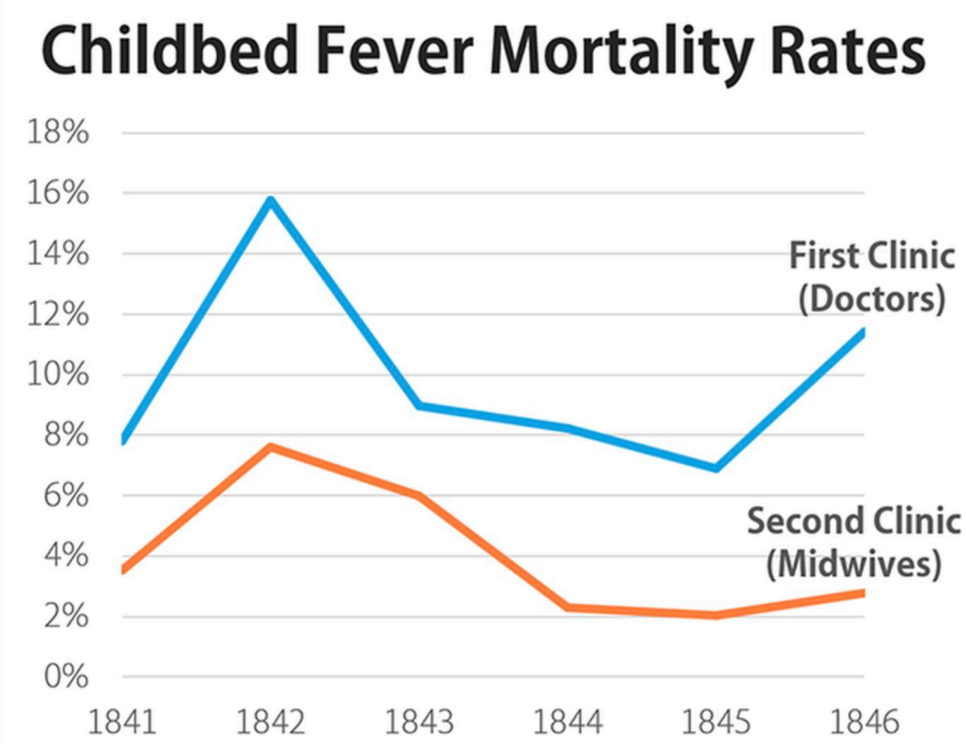
Source: https://www.forbes.com/sites/brentdykes/2016/02/09/a-history-lesson-on-the-dangers-of-letting-data-speak-for-itself/?sh=57d2e24820e1

Question

What was wrong with Ignaz Semmelweis' tabular approach to explain his theory?

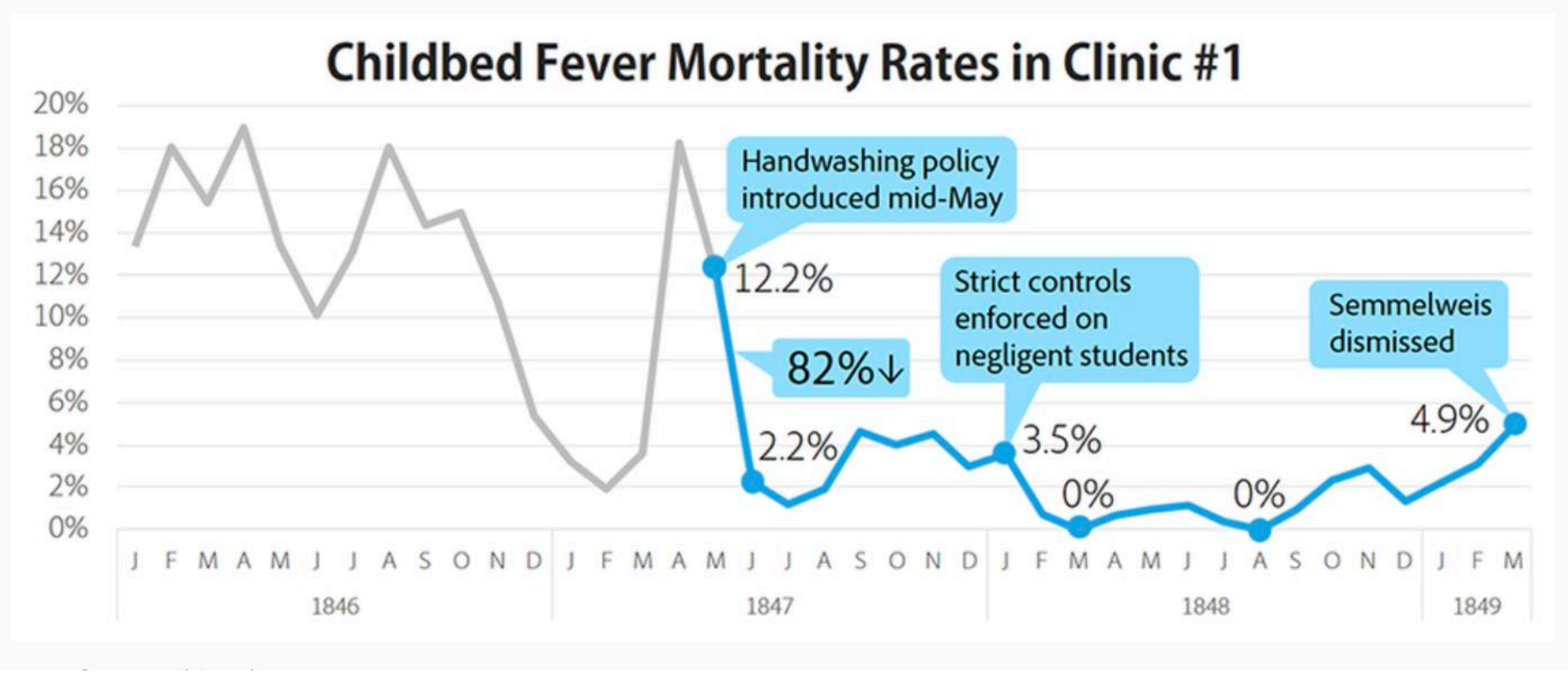
Source: https://www.forbes.com/sites/brentdykes/2016/02/09/a-history-lesson-on-the-dangers-of-letting-data-speak-for-itself/?sh=57d2e24820e1

What could he have been done. Simple line graph



Source: https://www.forbes.com/sites/brentdykes/2016/02/09/a-history-lesson-on-the-dangers-of-letting-data-speak-for-itself/?sh=57d2e24820e1

Line graph with monthly breakdown and callouts



Source: https://www.forbes.com/sites/brentdykes/2016/02/09/a-history-lesson-on-the-dangers-of-letting-data-speak-for-itself/?sh=57d2e24820e1

Statistics provide visibility; Narratives drive action

“All great truths begin as blasphemies.”

George Bernard Shaw

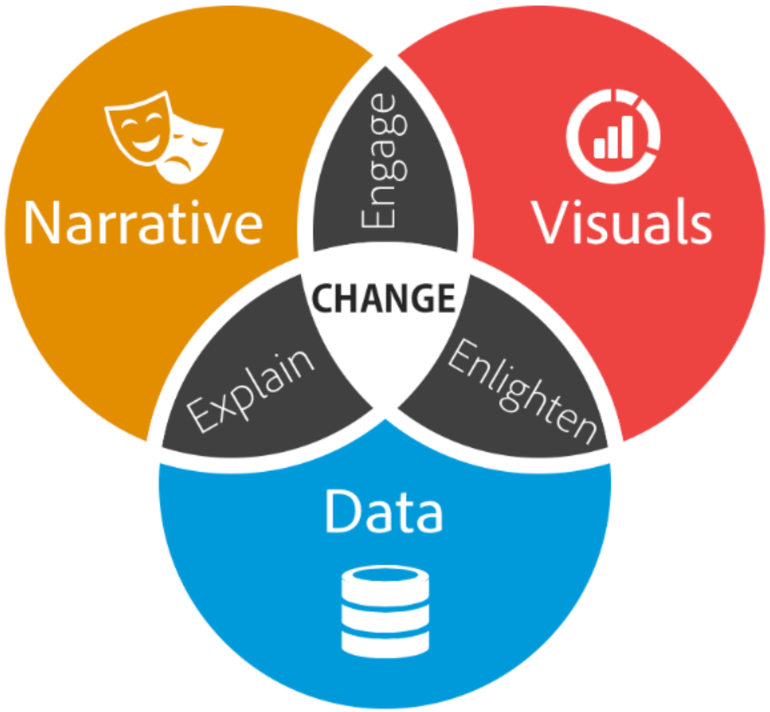
Source: https://www.forbes.com/sites/brentdykes/2016/02/09/a-history-lesson-on-the-dangers-of-letting-data-speak-for-itself/?sh=57d2e24820e1

Newtown's First Law of Motion?

As mathematician **John Allen Paulos** observed, "In listening to **stories** we tend to **suspend disbelief** in order to be **entertained**, whereas in evaluating **statistics** we generally have an opposite inclination to **suspend belief** in order not to be **beguiled**."

Explain & Enlighten & Engage → Change/Action

Storytelling with Data



Source: https://www.forbes.com/sites/brentdykes/2016/03/31/data-storytelling-the-essential-data-science-skill-everyone-needs/#503dfa8552ad

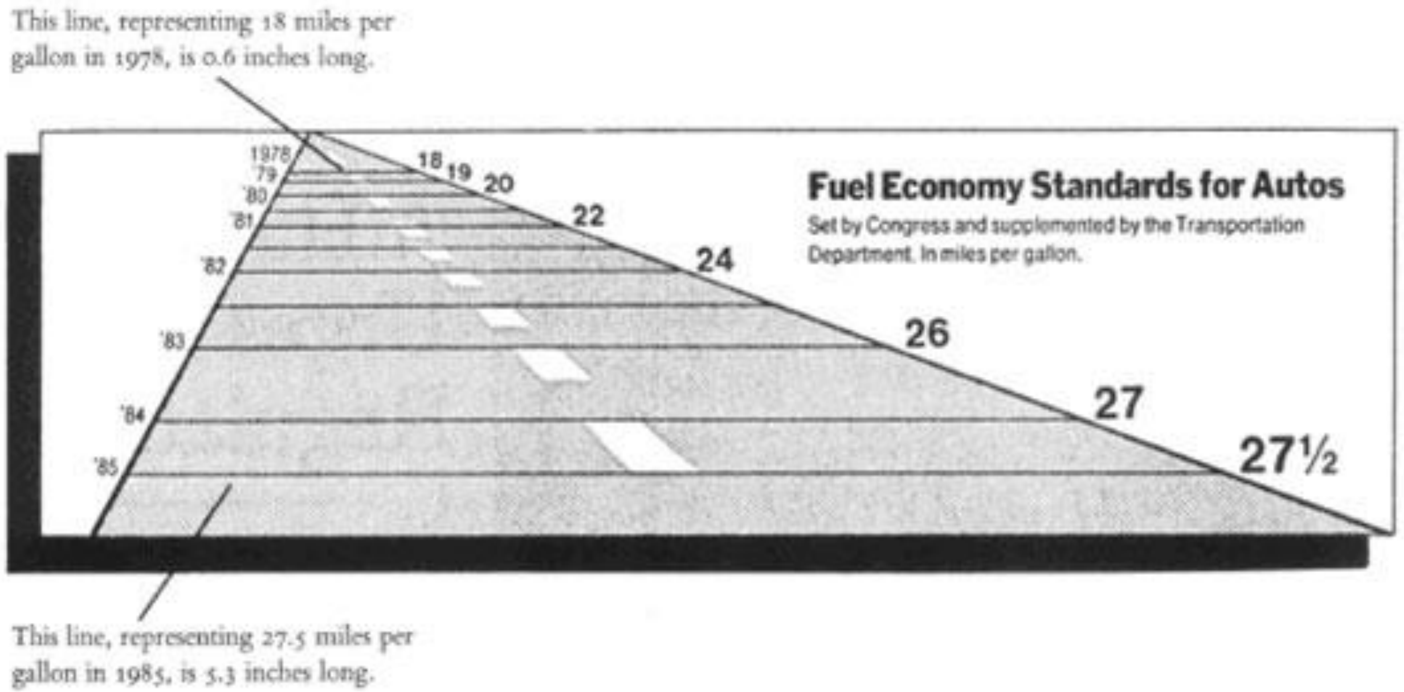
Key to good visualisations

Accuracy:
Represent data truthfully to avoid misleading interpretations.

Clarity:
Visuals should convey data clearly without clutter.

Focus:
Direct viewers' attention to the most important aspects of the data.

Bad Visualisation

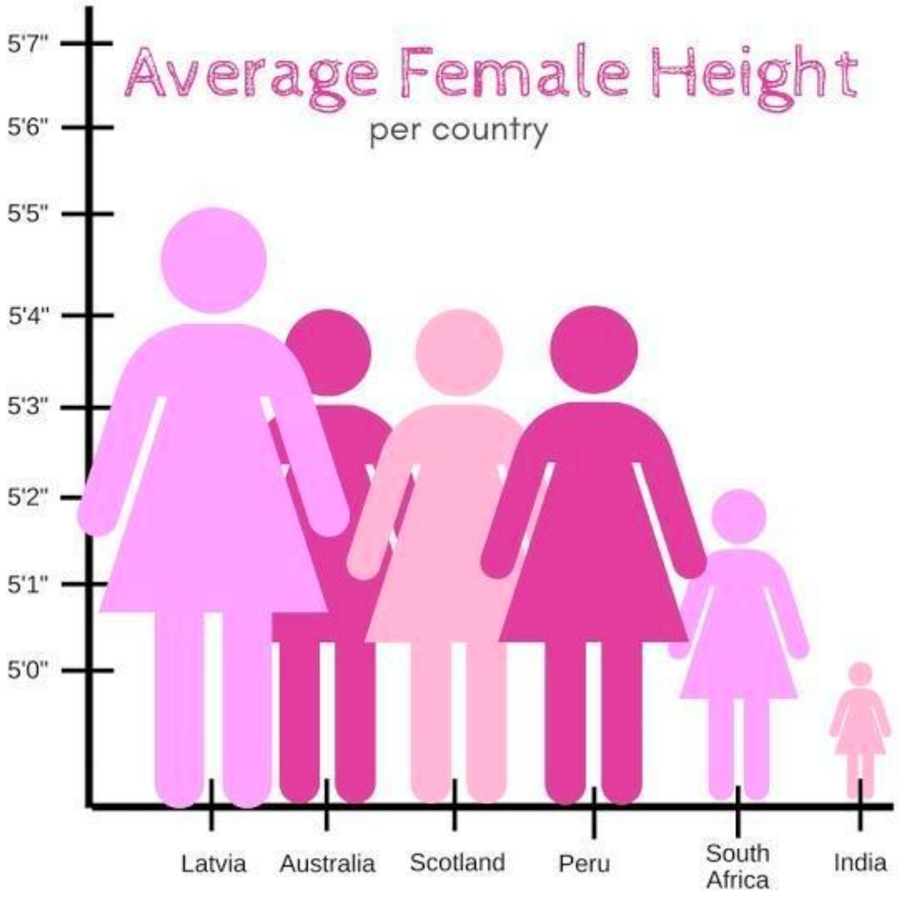


Mandated fuel economy standards for autos set by the US Department of Transportation

Critique: the standard required an increase in mileage from 18 to 27.5, an increase of 53%. The magnitude of increase shown in the graph is 783%, which results in a lie factor of 14.8! [Tufte, 1983, p.57]

$$\textit{Lie Factor} = \frac{\text{size of effect shown in graphic}}{\text{size of effect in data}} \qquad \textit{size of effect} = \frac{|\text{second value - first value}|}{\text{first value}}$$

Poor Visualisations

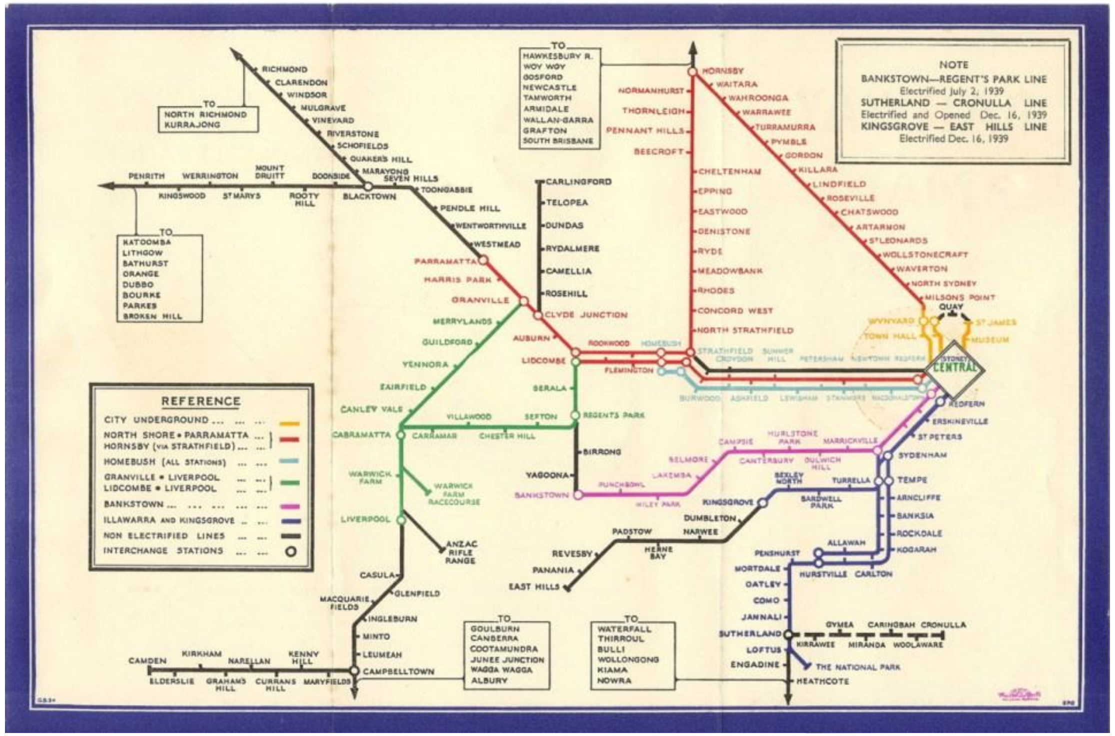


- Not starting with 0 makes it seem like Indian women are one-fifth the size of Latvian women
- Why are the figures scaled on the X-axis as well? Adds no information, creates confusion
- What do the colour tones represent?
- Used silhouette in place of bars to represent female. This can still be achieved with a bar chart and a change in title.
- Country flags can help with recall.

Source: https://badvisualisations.tumblr.com/

Good Visualisation – Historical Milestone

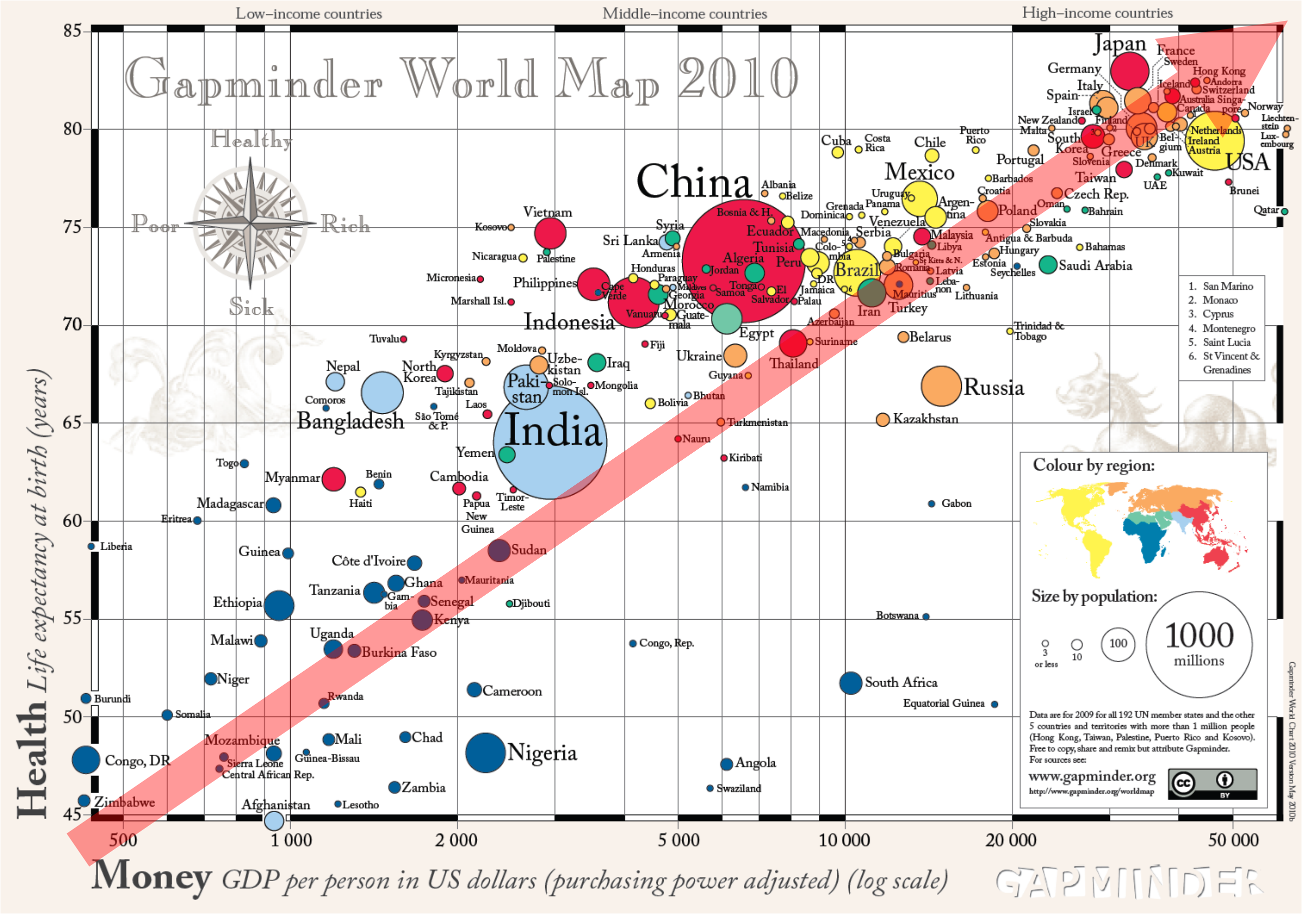
Sydney Real Geometrical Map



Sydney City-Rail Map

Source: https://transitmap.net/sydney-1939/

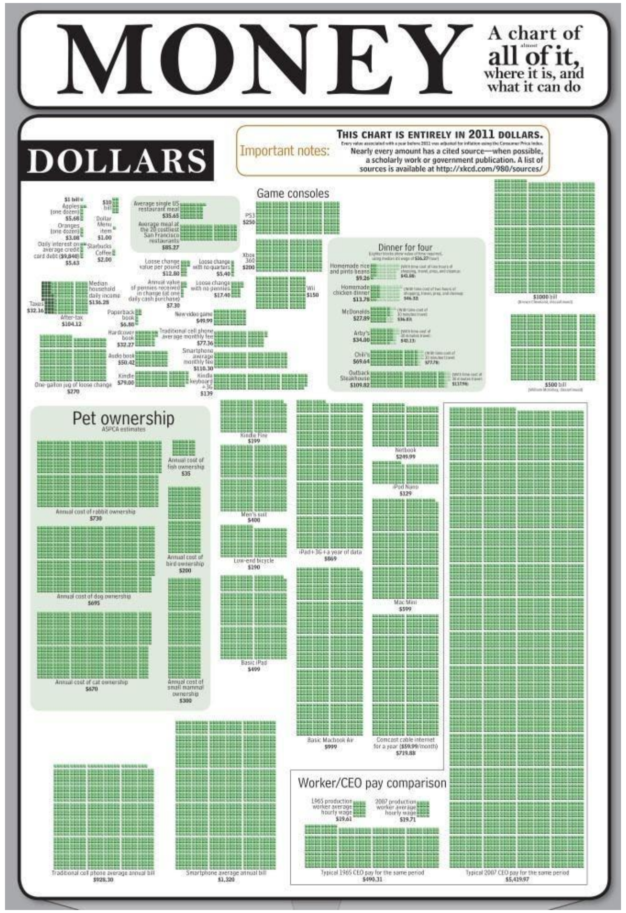
Good Visualisation – Colour, Size, Labels



Does adding an Arrow helps?

Source: https://www.fastcompany.com/1656059/higher-gdp-equals-longer-life-heres-chart-prove-it

Good Visualisation – Empathy and Relativity



Money: A chart of (almost) all of it, where it is, can what it can do.

Source: https://www.forbes.com/sites/davidleinweber/2012/12/31/the-mother-of-all-money-charts/

In data viz, start with getting the context right

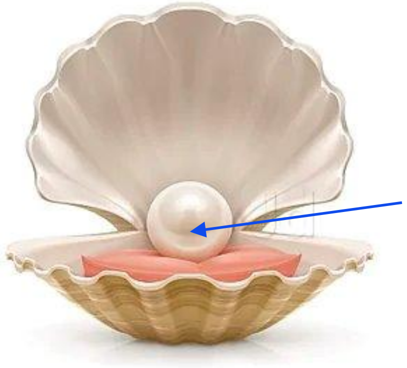
WHO (Audience)	WHAT (Message)	HOW (Design)
Tailor visuals to the audience's background and goals, adjusting complexity and focus accordingly.	Clarify the main insight or story, emphasizing key trends or comparisons relevant to the purpose.	Use the appropriate visualisation types, design elements, and tools to effectively communicate the data and message.

Decide on Exploratory or Explanatory visuals

Exploratory:
Open-ended insights, interactive, used in analysis phases.

Hunting for pearls in clams

Explanatory:
Directed insights, specific audience, static visuals.



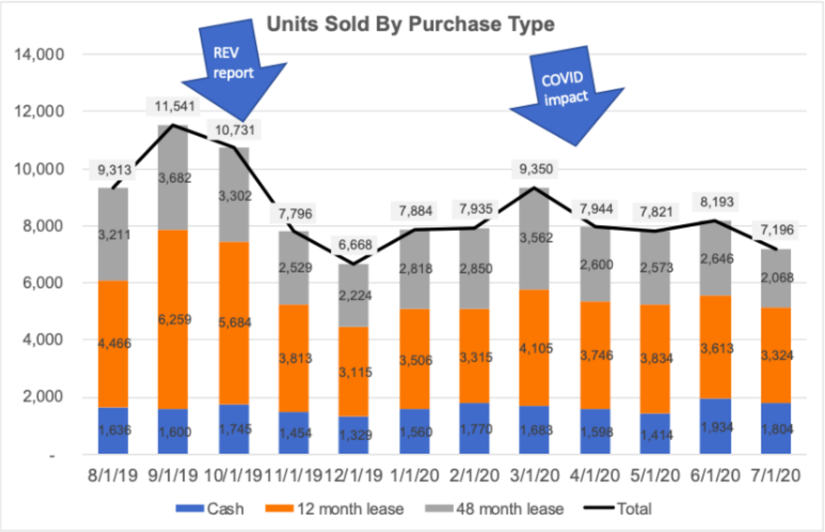
Key message

Communicating a specific story

Idea source: Joanna Young and Jan Wessnitzer. 2016. Descriptive statistics, graphs, and visualisation. In *Modern statistical methods for HCI*, Kaptein M Robertson J. (Ed.), Springer, Cham, 37-56. https://link-springer-com.ezproxy.lib.uts.edu.au/chapter/10.1007/978-3-319-26633-6_3

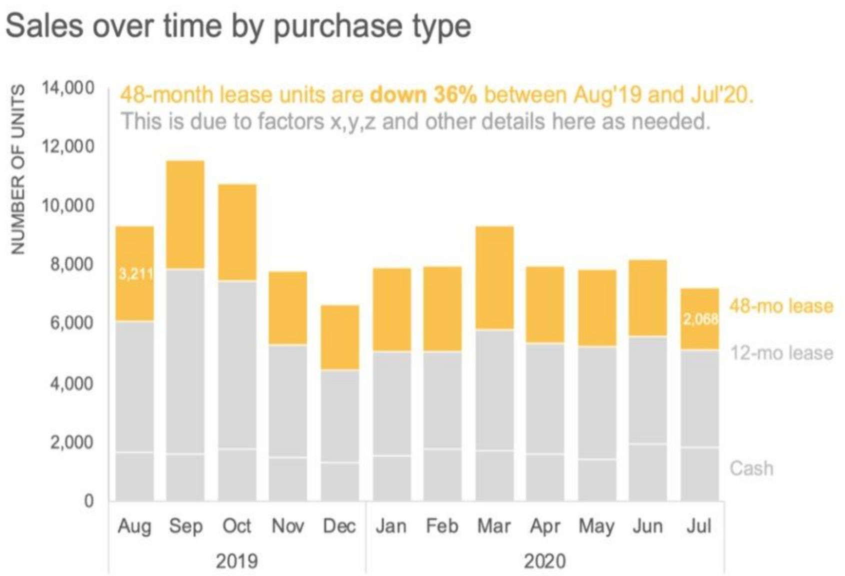
Exploratory vs Explanatory

Hunting for pearls in oysters



ANALYSIS FINDINGS: Total unit decline in 12-month (-26%) and 48-month (-36%) from August to July, while cash grew (10%).

Communicating a specific story



Exploratory visualisation - Theory

Purpose:
Enables data exploration and discovery, often by analysts.

Audience:
Typically used by data scientists or analysts to find patterns, trends, or anomalies.

Examples:
Scatter plots, heat maps, interactive dashboards.

Exploratory visualisation - examples

- Convolutional Neural Network Explorer
- Social Network Analysis
- LinkedIn Connections Between J and K

Hunting for pearls in oysters

Explanatory visualisation - Theory

Purpose:
Designed to convey a clear, specific message or insight.

Audience:
Often for presentations, reports, and decision-making; focused on storytelling.

Examples:
Pie charts showing market share, line graphs showing trends.

Explanatory visualization - goals

Communicating insights

Telling the story with visuals

Helps recipient make evidence-based decisions

But… why do we use visualisations?

Visual reasoning is way faster and more reliable than mental reasoning.

External Cognition

Why visualisation?

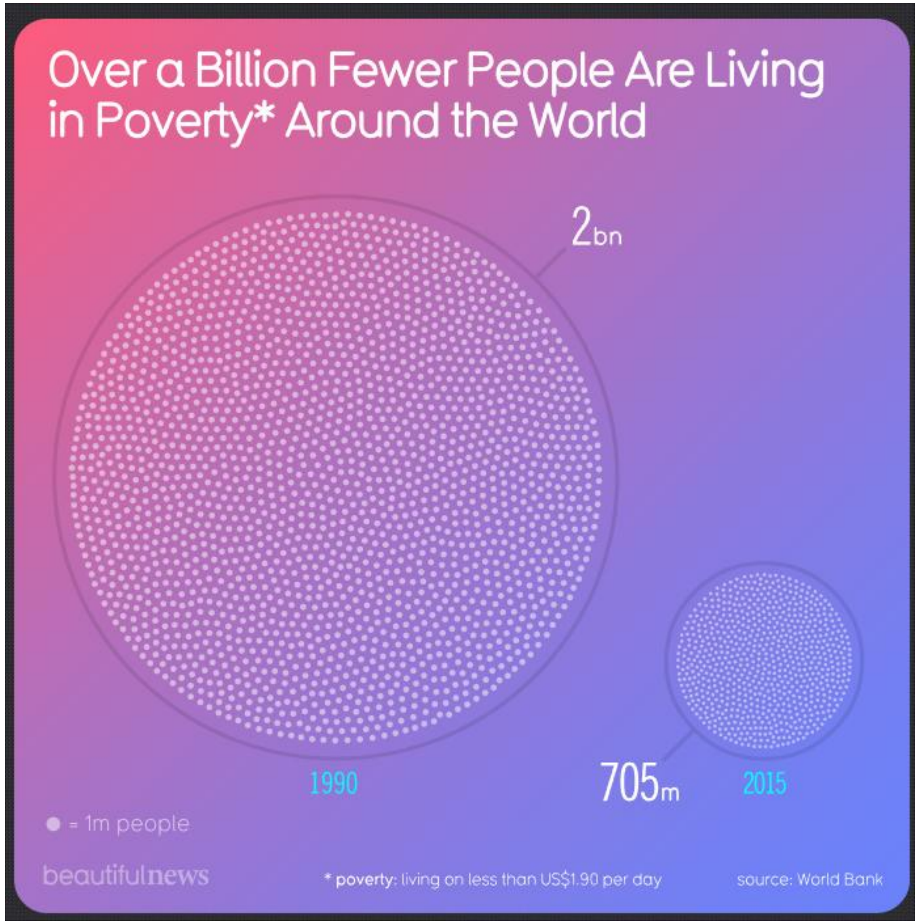
- Could visualisation amplify cognition/perception?
- Could visualisation help to solve daily problems?
- Could visualisation be useful for big data analysis?

"A good picture is worth a thousand words!"

Use a picture to tell a story instead of a report full of text:

https://informationisbeautiful.net/beautifulnews/

https://www.voronoiapp.com/



Visual perception

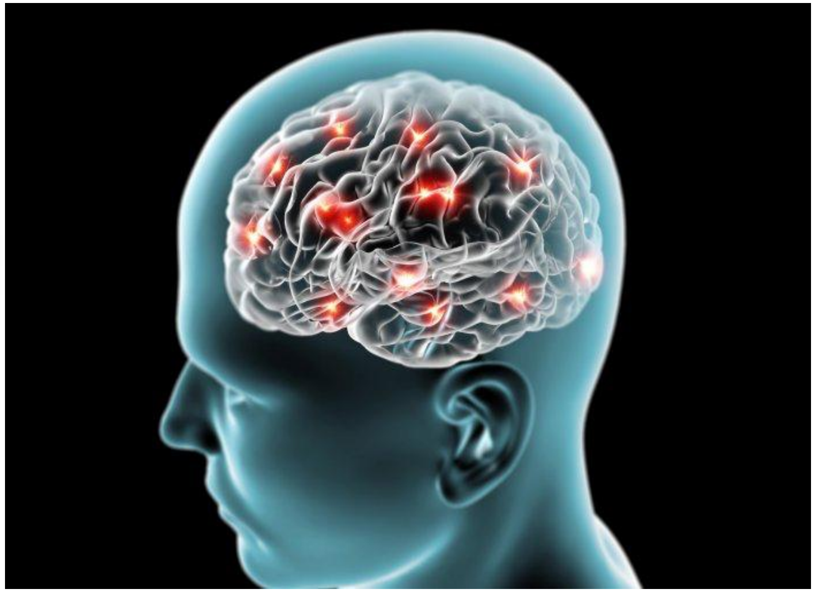
Definition:
Visual perception is the ability to interpret surroundings using light that reaches the eyes, forming images in the brain.

Importance:
Critical for understanding and interacting with the environment.

Key Processes:
Includes processes like recognition, depth perception, and motion detection.

Visual perception

Stimulus → Eye → Brain



Let’s do a simple exercise to validate how our visual reasoning is faster than mental reasoning

An exercise in pre-attentive processing

56254982354593
56849902765409
09374176352638
87246252835912

How many number 3 appears in the sequence?

An exercise in pre-attentive processing

56254982354593

56849902765409

09374176352638

87246252835912

Understanding visual pathway

Light enters through the cornea and lens and focuses on the retina.

Photoreceptor cells in the retina (rods and cones) detect light and color.

Optic nerve transmits signals to the brain.

Visual cortex located in the brain's occipital lobe interprets these signals as images.

Components of visual perception

1

Depth Perception: Allows us to judge distances and perceive the world in three dimensions.

2

Color Perception: Cones in the retina enable color differentiation.

3

Motion Detection: Detects movement, crucial for survival and navigation.

4

Object Recognition: Ability to recognise shapes and patterns.

Factors influencing visual perception

Lighting Conditions:
Affects clarity and color.

Contrast and Color:
High contrast improves visibility; certain colors are perceived faster.

Experience and Memory:
Past experiences and memory shape perception.

Optical Illusions:
Demonstrates how perception can be deceived.

Applications of visual perception in technology

Computer Vision: Mimics human visual perception in object detection and facial recognition.

Virtual Reality (VR): Enhances immersive experiences through accurate depth and motion perception.

Assistive Devices: Use visual cues to aid those with vision impairments.

Image from: https://datavizcatalogue.com/

Section 3

Choosing an Appropriate Visualisation

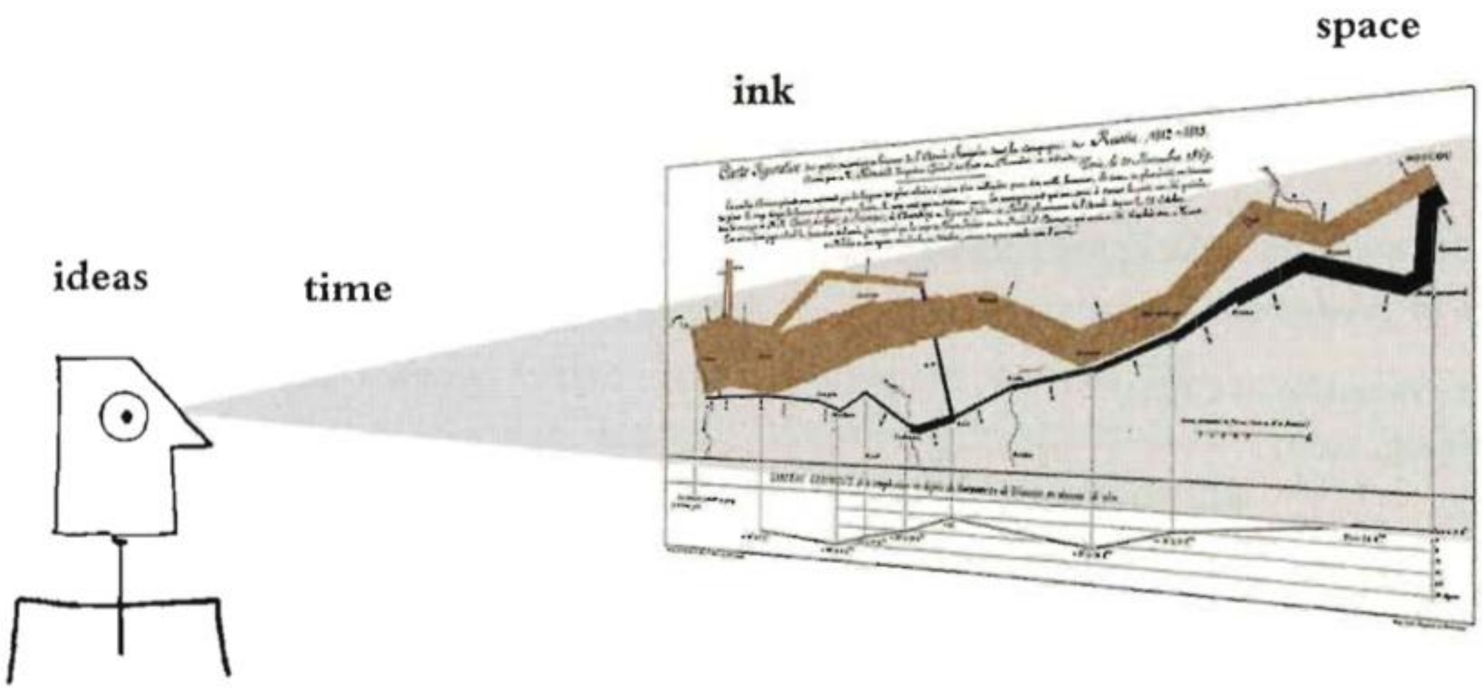
Edward Tufte’s framework – Graphical Integrity

Principals of Analytical Design

1. **Comparisons:** Use visual comparisons, like bar charts, to highlight the differences between key data points.
2. **Causality:** Show the impact of independent variables on the outcomes.
3. **Multivariate:** Present a comprehensive view by integrating diverse data points for easy interpretation.
4. **Integration:** Combine different forms of information, such as text, maps, calculations and visuals, to provide a complete picture of the data and its origin
5. **Documentation:** Enhance credibility by providing clear citations, detailed labels, and measurement scales.
6. **Context:** Give perspective by illustrating the data's progression and hinting at potential future developments through trend analysis.

Edward Tufte's framework – Data-ink Ratio

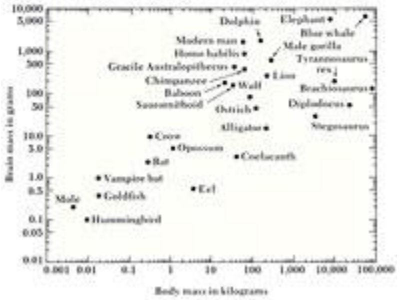
- Maximise the amount of ink used to represent data
- Minimise the amount of ink used to represent non-data elements
- The ideal visual has the highest possible data-ink ratio

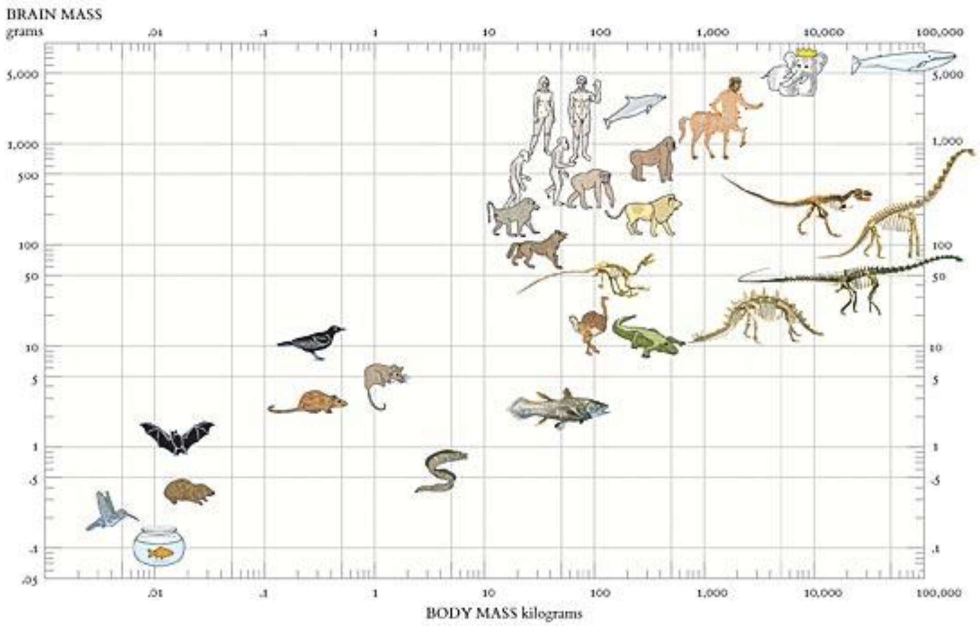


Refection Activity: https://canvas.uts.edu.au/courses/34231/pages/principles-of-graphical-excellence?module_item_id=1959546
Learn more about Minard's Map here: https://www.esri.com/content/dam/esrisites/en-us/esri-press/book-pages/sample-page/mapping-time-illustrated-minards-map-napoleons-russian-campaign-1812.pdf

Edward Tufte’s framework – Visual Excellence

- Give the viewer the greatest number of ideas in the shortest time
- Use the least amount of ink in the smallest space
- Strive for high-quality design in visualisations
- Ensure visual representations are truthful and clear





Source: https://nymag.com/arts/books/features/33156/

Understanding what is available

Visuals selected must effectively communicate complex information.

Edward Tufte identified various data visualisation types, each suited for specific data contexts.

Examples: Maps, time-series, space-time, narrative design, relational graphs, and others.

Search a catalogue: https://datavizcatalogue.com/, https://altair-viz.github.io/gallery/index.html#example-gallery, https://www.tableau.com/solutions/gallery/visual-vocabulary

Use a decision tree: https://www.data-to-viz.com/

Read a guide: https://www.atlassian.com/data/charts/how-to-choose-data-visualization

Be inspired: https://public.tableau.com/app/discover

Understand the terminologies and features

Table

Slopegraph

Timeline

Simple Text

Line

Stacked horizontal bar

Heatmap

Scatterplot

Graphs

Square area

Vertical graph

Waterfall

Other complex graphs

Stacked vertical bar

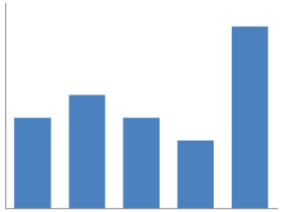
Horizontal graph

Commonly used charts

91%

Simple text

Scatterplot



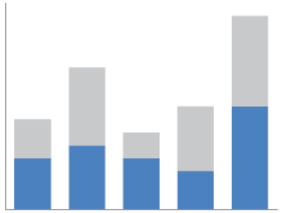
Vertical bar

Horizontal bar

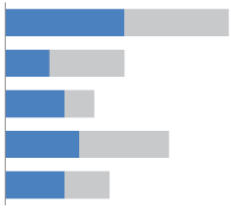
	A	B	C
Category 1	15%	22%	42%
Category 2	40%	36%	20%
Category 3	35%	17%	34%
Category 4	30%	29%	26%
Category 5	55%	30%	58%
Category 6	11%	25%	49%

Table

Line



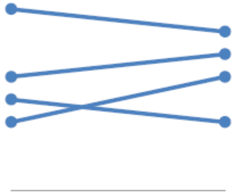
Stacked vertical bar



Stacked horizontal bar

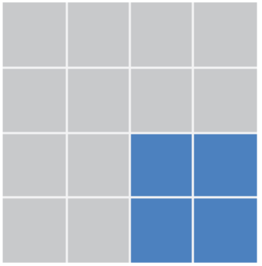
	A	B	C
Category 1	15%	22%	42%
Category 2	40%	36%	20%
Category 3	35%	17%	34%
Category 4	30%	29%	26%
Category 5	55%	30%	58%
Category 6	11%	25%	49%

Heatmap



Slopegraph

Waterfall

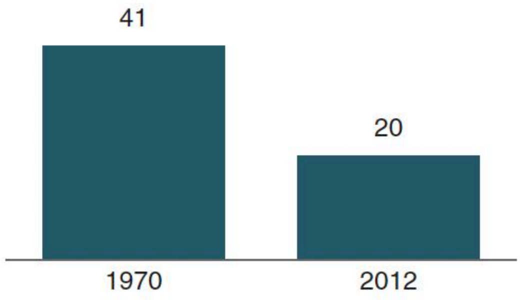


Square area

Simple text can be effective for explanatory purposes

Children with a "Traditional" Stay-at-Home Mother

% of children with a married stay-at-home mother with a working husband



Note: Based on children younger than 18. Their mothers are categorized based on employment status in 1970 and 2012.

Source: Pew Research Center analysis of March Current Population Surveys Integrated Public Use Microdata Series (IPUMS-CPS), 1971 and 2013

Adapted from PEW RESEARCH CENTER

20%

of children had a **traditional stay-at-home mom** in 2012, compared to 41% in 1970

Tables aid exploratory efforts, not great in live demos

Types of tables:

- ❑ Heavy borders
- ❑ Light borders
- ❑ Minimal borders

Heavy borders

Group	Metric A	Metric B	Metric C
Group 1	$X.X	Y%	Z,ZZZ
Group 2	$X.X	Y%	Z,ZZZ
Group 3	$X.X	Y%	Z,ZZZ
Group 4	$X.X	Y%	Z,ZZZ
Group 5	$X.X	Y%	Z,ZZZ

Light borders

Group	Metric A	Metric B	Metric C
Group 1	$X.X	Y%	Z,ZZZ
Group 2	$X.X	Y%	Z,ZZZ
Group 3	$X.X	Y%	Z,ZZZ
Group 4	$X.X	Y%	Z,ZZZ
Group 5	$X.X	Y%	Z,ZZZ

Minimal borders

Group	Metric A	Metric B	Metric C
Group 1	$X.X	Y%	Z,ZZZ
Group 2	$X.X	Y%	Z,ZZZ
Group 3	$X.X	Y%	Z,ZZZ
Group 4	$X.X	Y%	Z,ZZZ
Group 5	$X.X	Y%	Z,ZZZ

Heatmaps converted from tables are easier to digest

Table

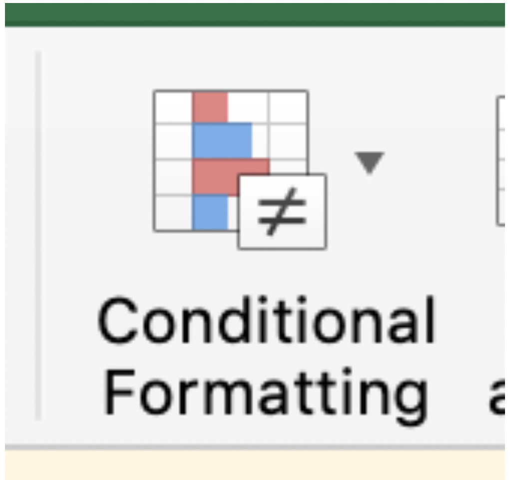
	A	B	C
Category 1	15%	22%	42%
Category 2	40%	36%	20%
Category 3	35%	17%	34%
Category 4	30%	29%	26%
Category 5	55%	30%	58%
Category 6	11%	25%	49%

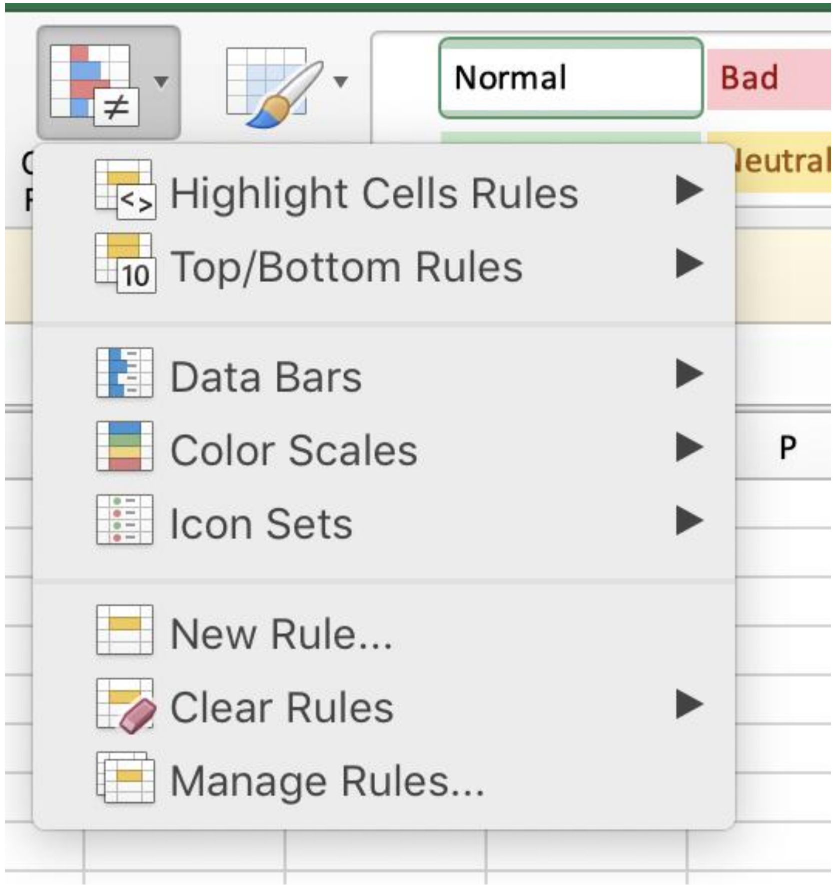
Heatmap

LOW-**HIGH**

	A	B	C
Category 1	15%	22%	42%
Category 2	40%	36%	20%
Category 3	35%	17%	34%
Category 4	30%	29%	26%
Category 5	55%	30%	58%
Category 6	11%	25%	49%

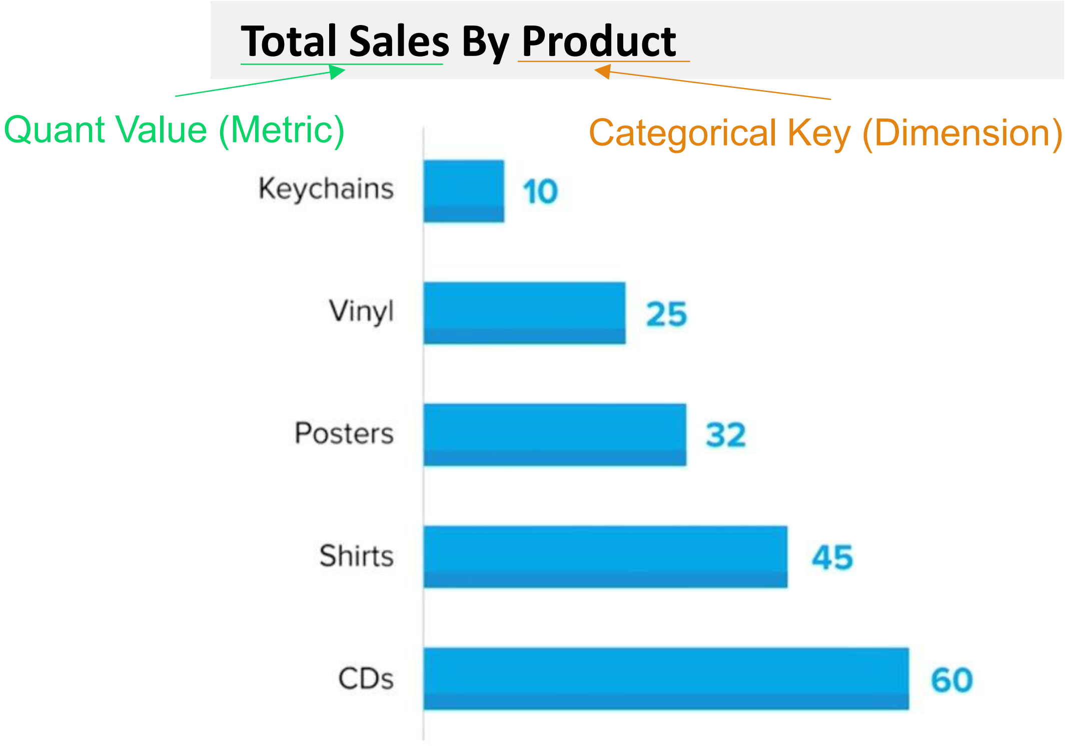
Heatmaps achieved using simple conditional formatting



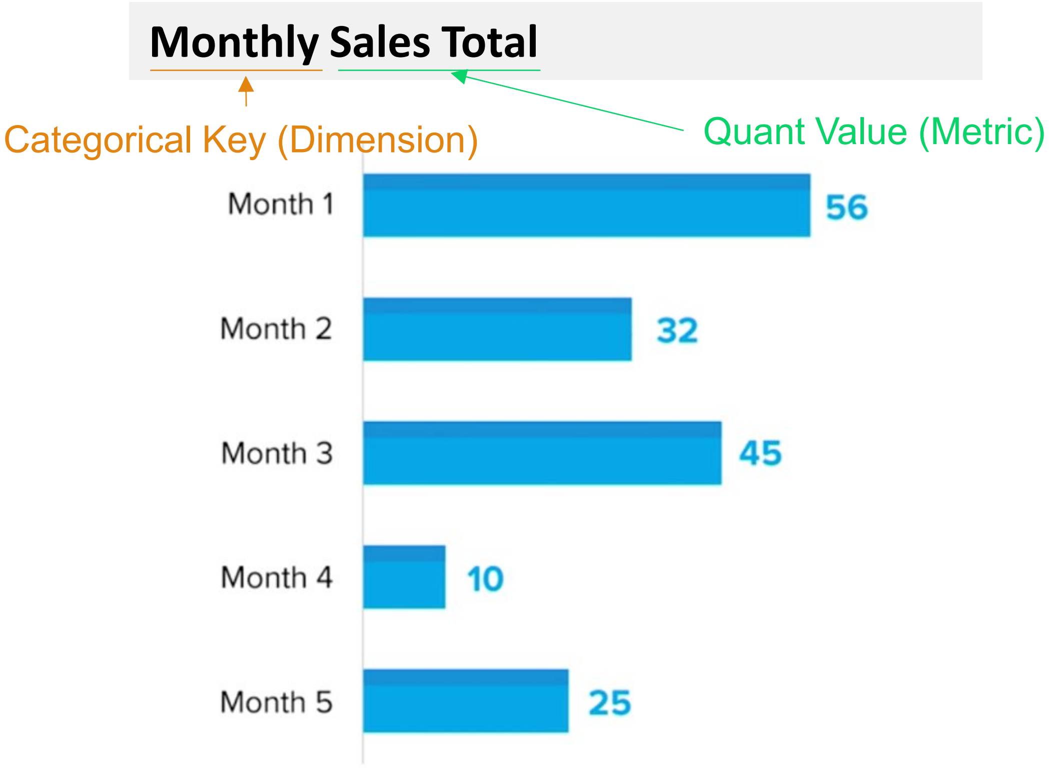


Bar chart: 1 categorical key, 1 quantitative value

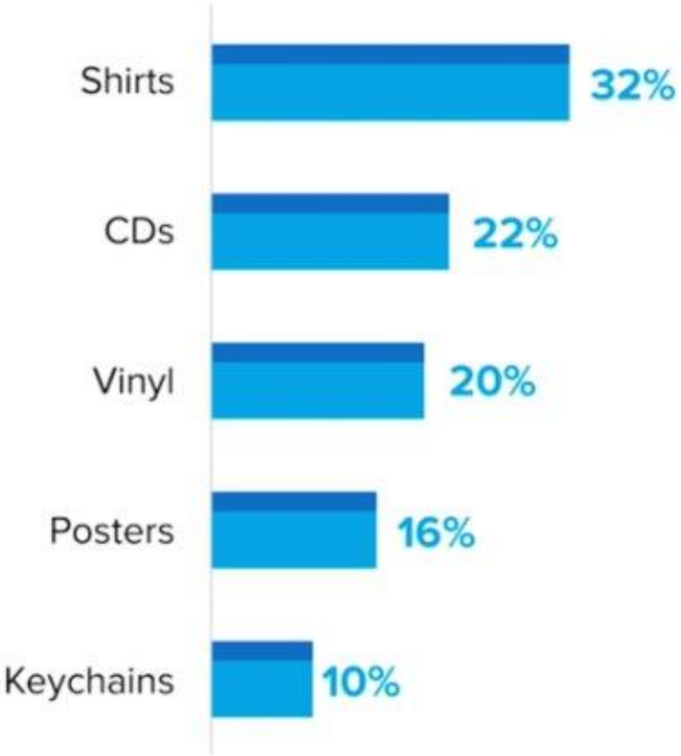
Tip 1: Organise from lower to higher metric value or vice versa (depending on the message)

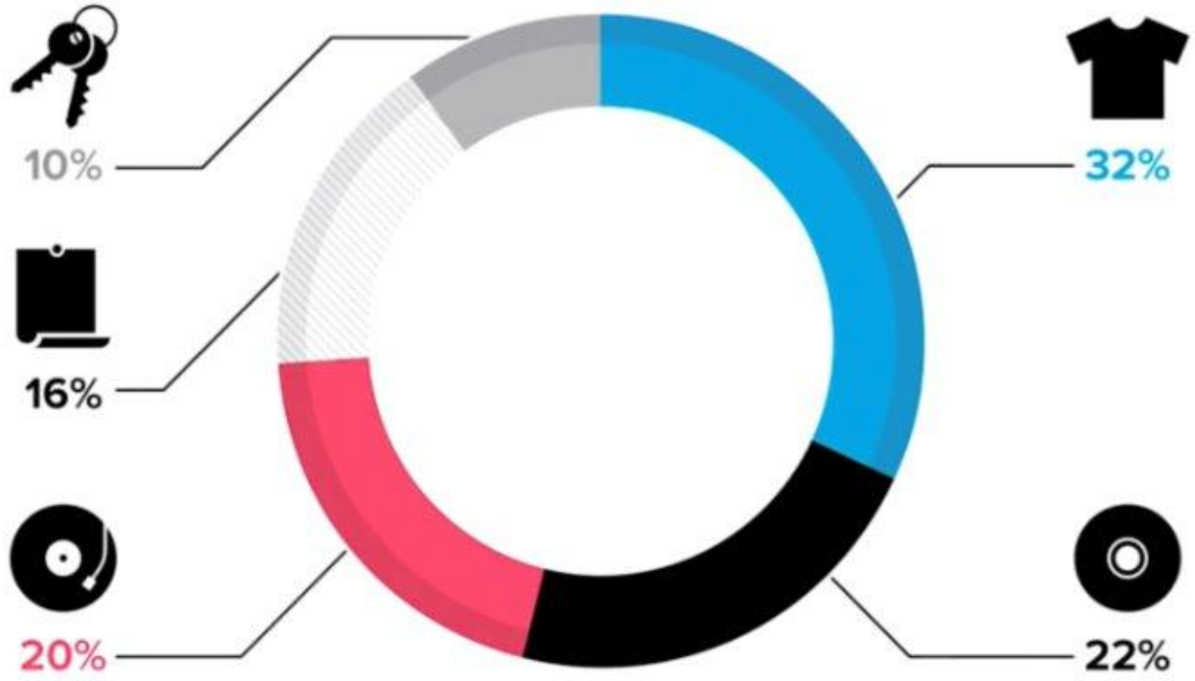


Tip 2: Ordered dimension provides continuity



Always consider different options for your audience





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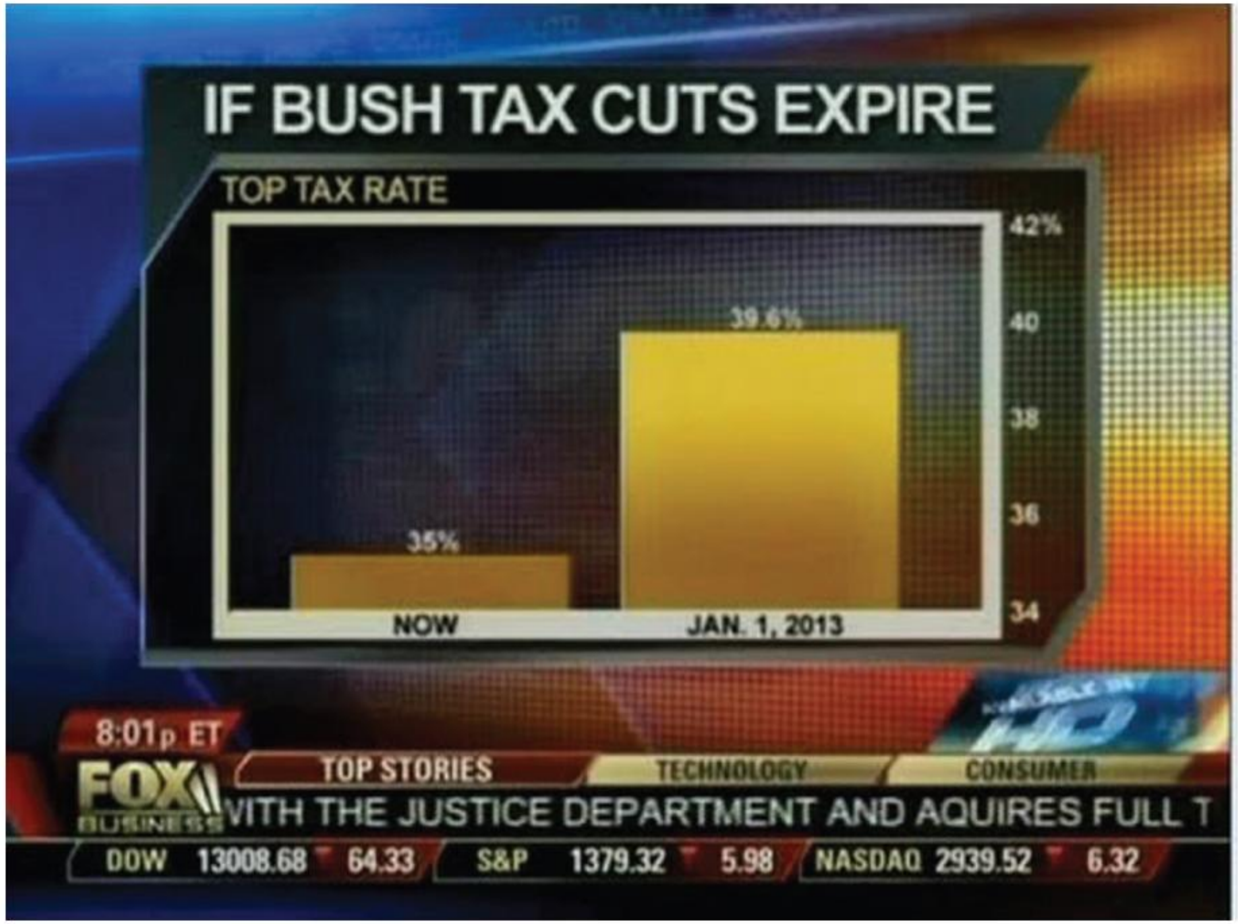
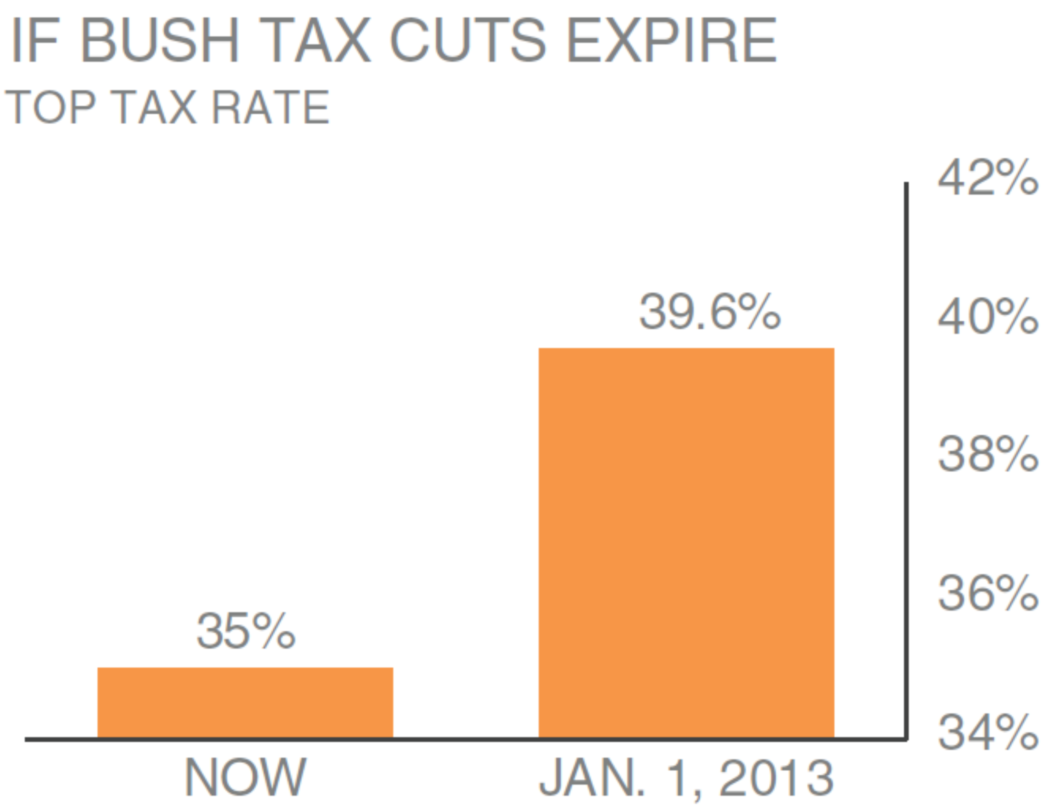
Bar chart in news: how does this make you feel?

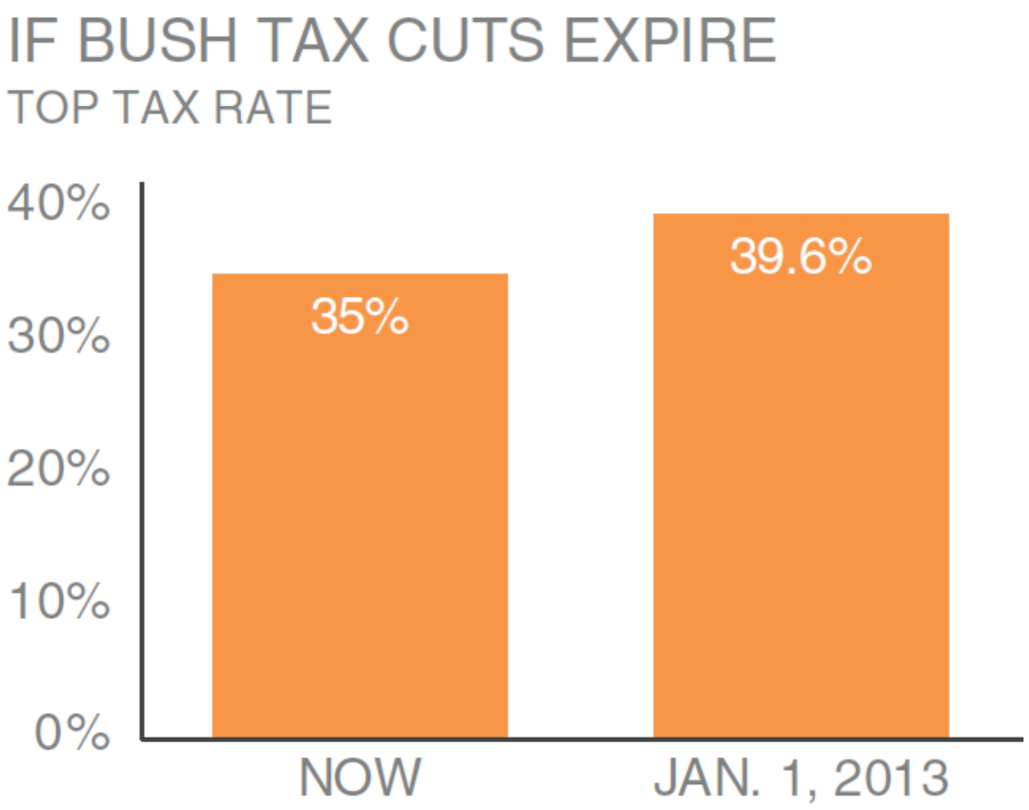


Bar Chart: With non-Zero baseline vs Corrected

Non-zero baseline: as originally graphed

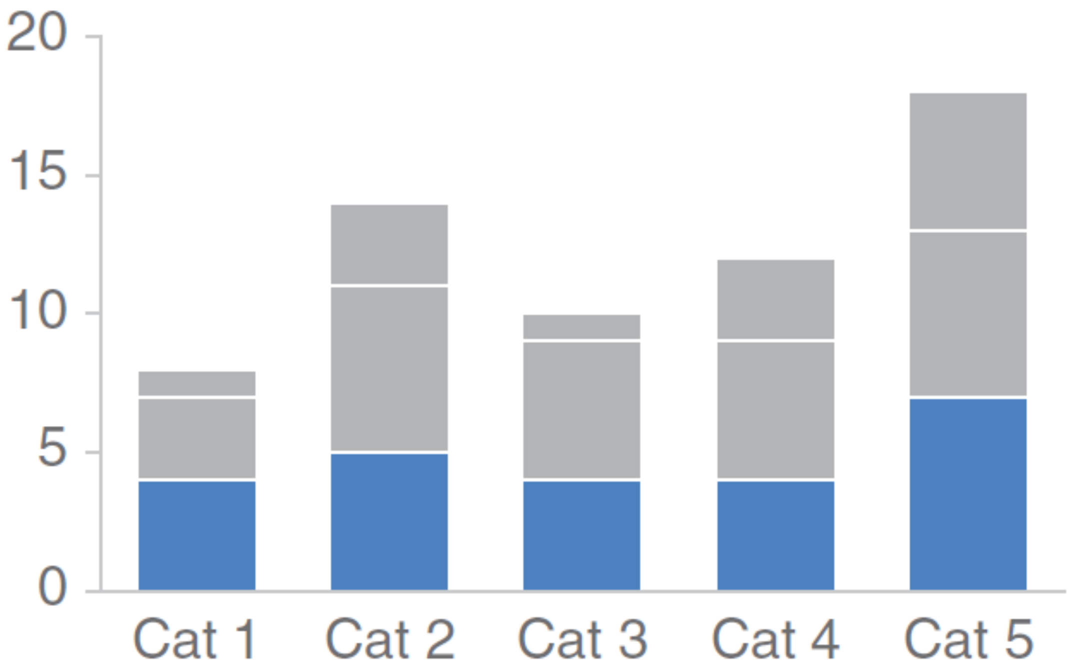


Zero baseline: as it should be graphed

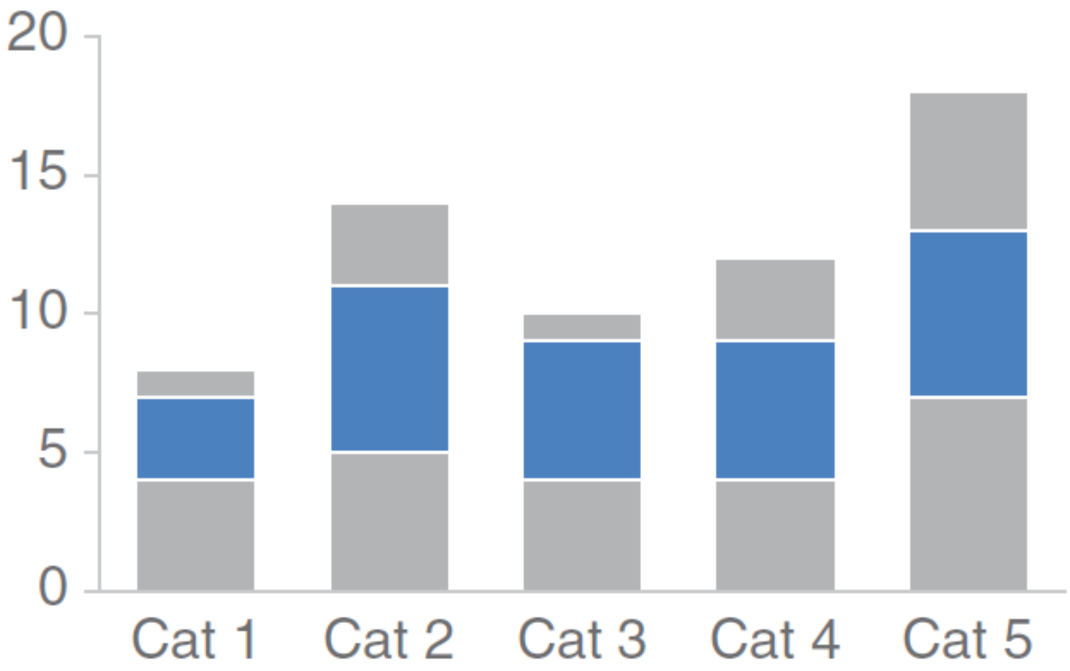


Stacked bar chart: understand pros and cons

Comparing these is easy



Comparing these is hard



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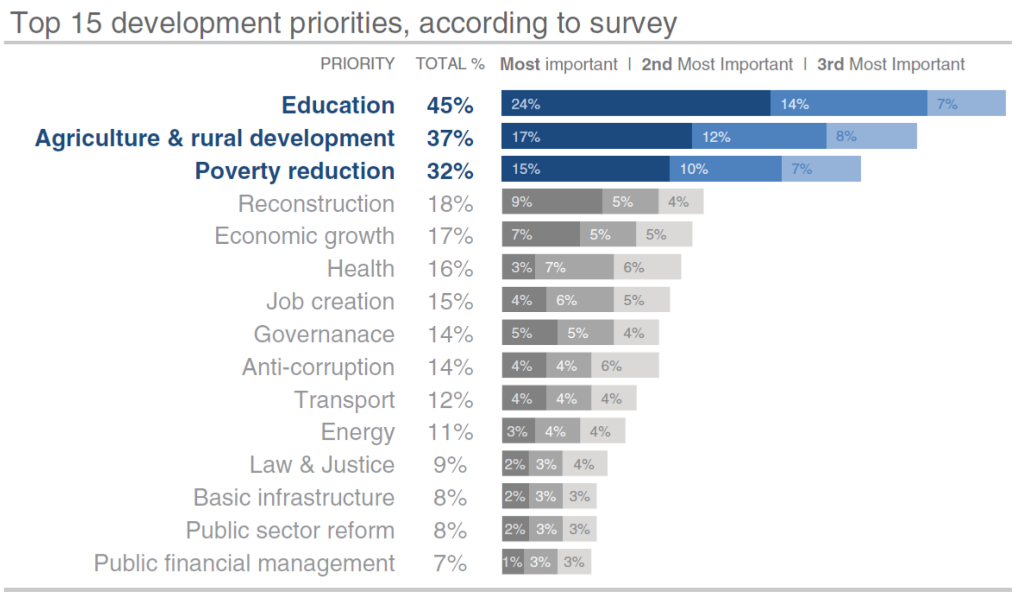
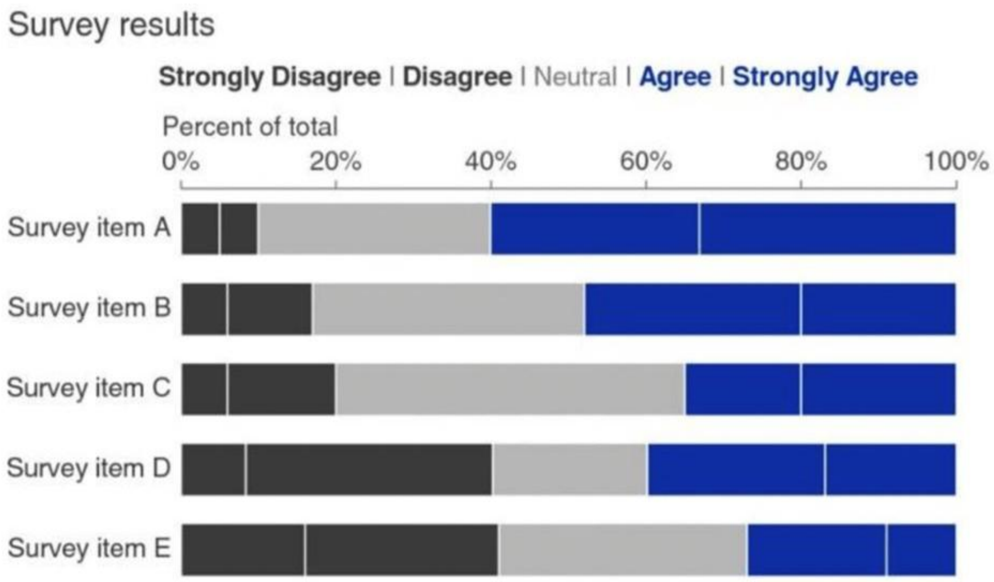
Horizontal stacked bar chart

Stacked Horizontal Bar Chart to focus on part-to-whole relationship



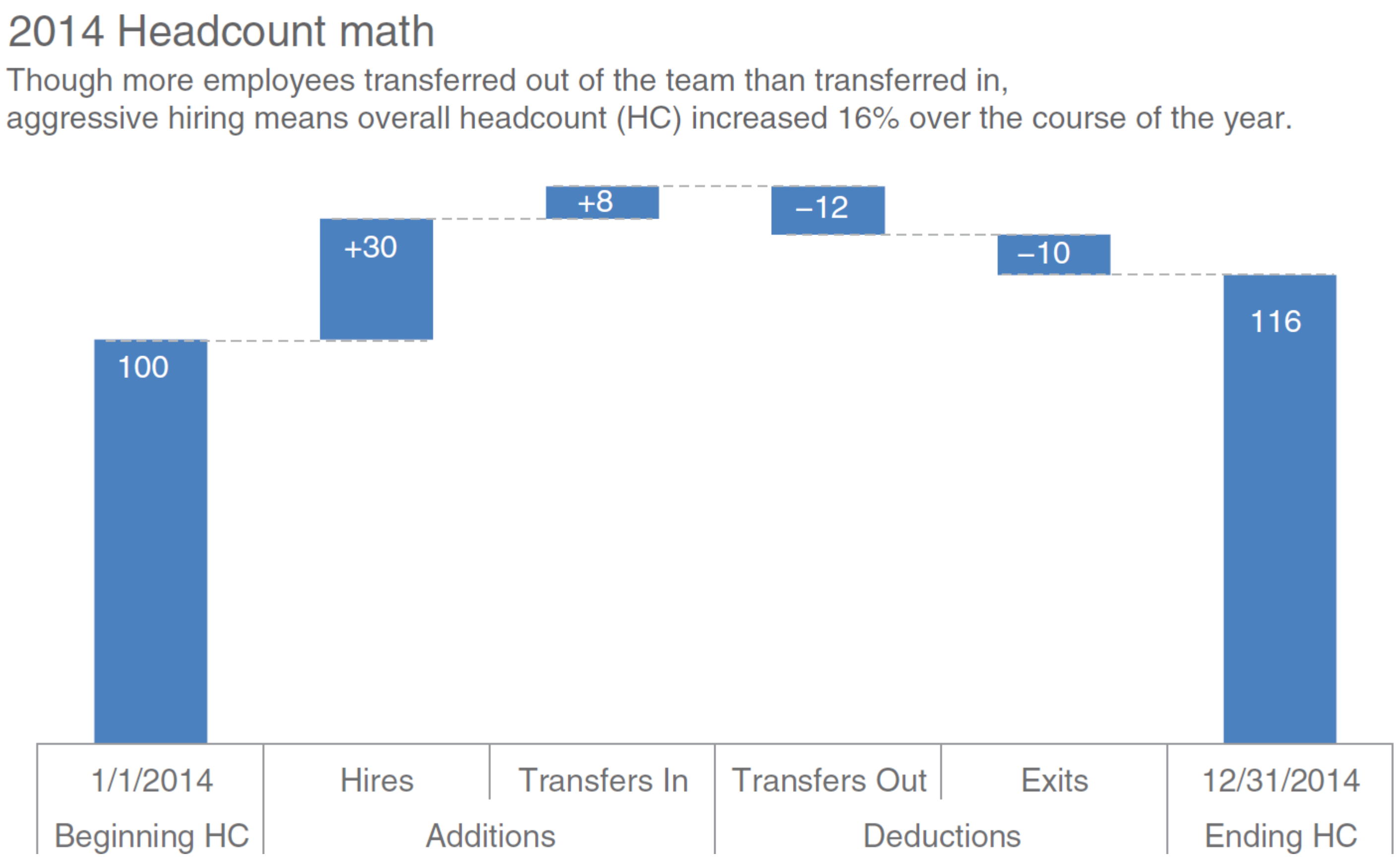
N = 4,392. Based on responses to item, *When considering development priorities, which one development priority is the most important? Which one is the second most important priority? Which one is the third most important priority?* Respondents chose from a list. Top 15 shown.

100% Stacked Horizontal Bar Chart focuses on proportions across categories for relative comparison



Waterfall chart helps to break down changes

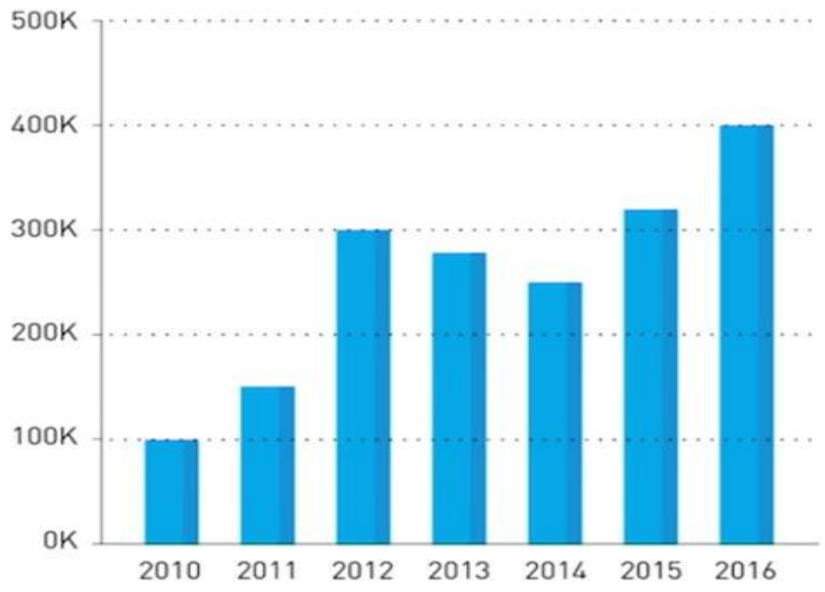
Imagine that you are an HR business partner and want to understand and communicate how employee headcount has changed over the past year for the client group you support.



Bar charts vs line graphs

Bar Charts

- Comparing categories or groups.
- Displaying discrete data, often in categorical or ordinal form.

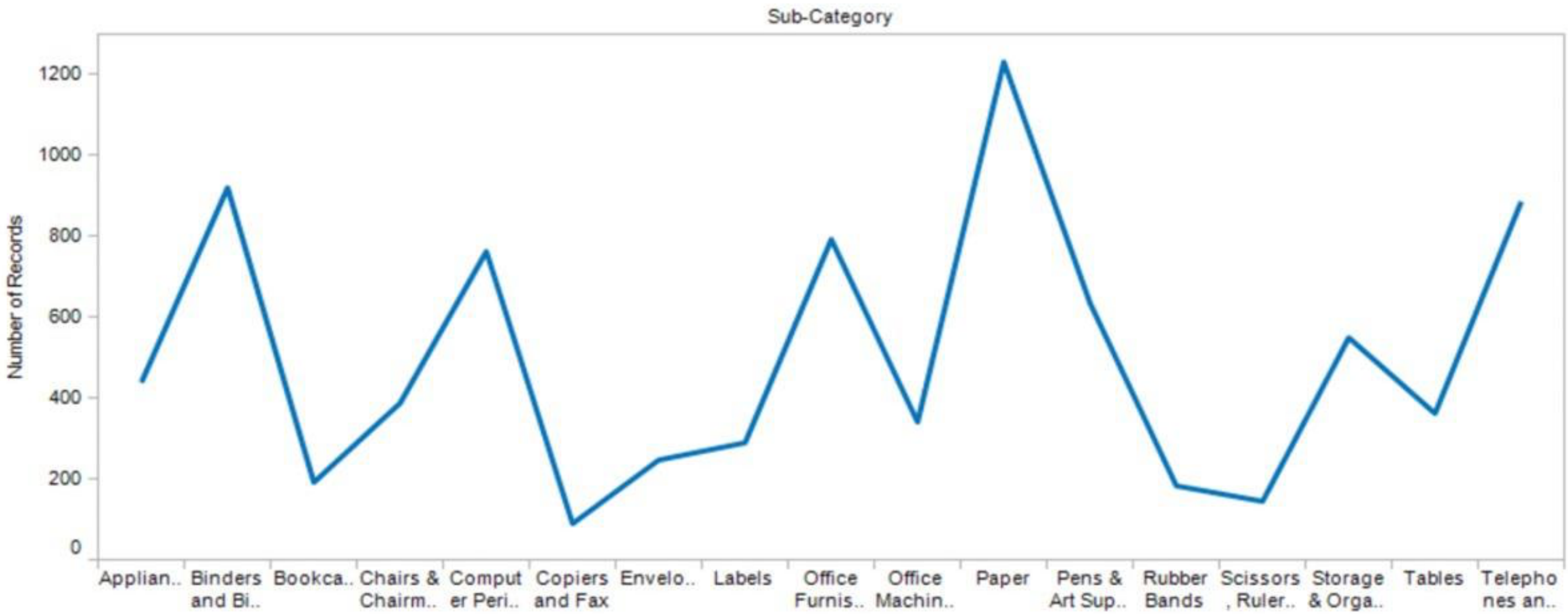


Line Graphs

- Showing trends over time.
- Displaying continuous data.
- Tracking changes and progressions across a timeline.

Why do we use a line to connect values?

Line chart with categorical data (Wrong!)



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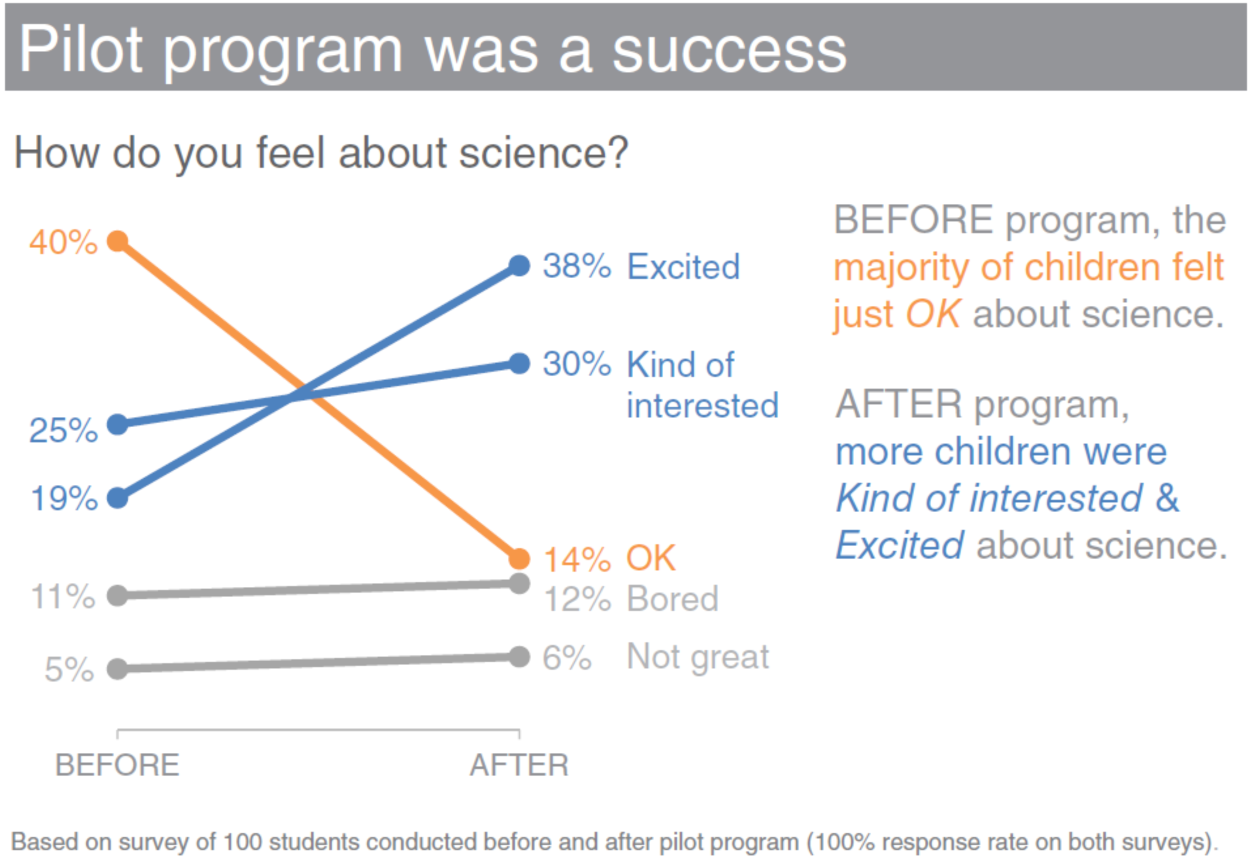
Slopegraph

Compares the “before” and “after states of different categories or items

Easy to Understand: Simple, intuitive and have broad appeal; Slope Direction, Gradient and Line Crossing provides meaning.

Focus on Change: Quickly draws attention to the most significant changes.

Space Efficient: Can display a reasonable amount of data in a compact format.



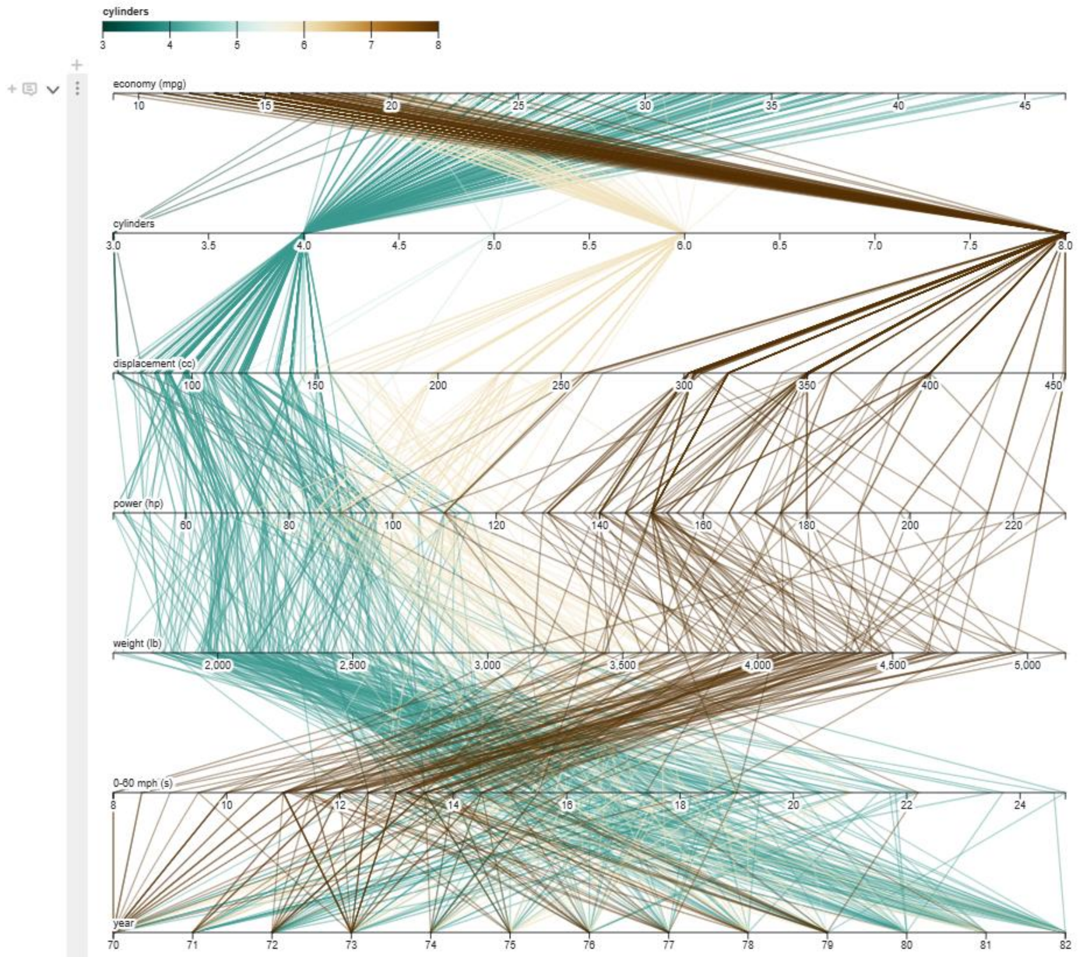
Practice here: https://www.storytellingwithdata.com/blog/2013/11/slopegraph-template

Parallel Coordinates: Useful As An Exploratory Tool

Fantastic way to visualize and explore relationships in multivariate data

Highlights: Cars with more cylinders have lower fuel economy, are generally more powerful, is heavier and get to 0-60 mph faster

Source: https://observablehq.com/@d3/parallel-coordinates



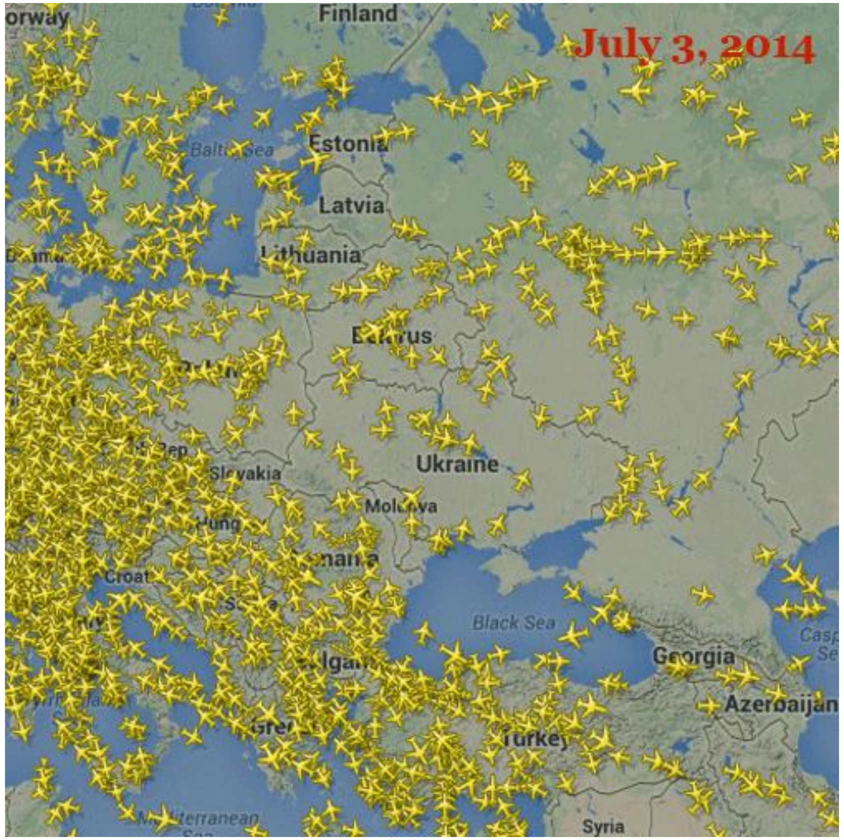
Space-Time Visuals

Visualize data across both spatial and temporal dimensions.

Uses: Tracking moving entities (e.g. closure of flight paths, hand and finger positions).

Characteristics: Typically use dynamic visuals like animations or 3D models.

Source: https://www.8thwall.com/8thwall/hand-tracking-threejs



Source: https://time.com/3002807/malaysia-airlines-ukraine-crash-airspace/

Quantagrams

Repeated pictograms or icons to represent a quantity

Keep it simple:
Choose clear and easily recognizable symbols.

Provide a key:
Explain what each symbol represents.

Don't overcrowd:
Use them for small to medium-sized datasets.

1. Simple Pictogram Quantagram:

🍎 = 5 people

- 🍎🍎🍎 = 15 people (Apples)
- 🍌🍌 = 10 people (Bananas)
- 🍇🍇🍇🍇 = 20 people (Grapes)

2. Proportional Symbol Size

- 👤 = 100 visitors
- 👥 = 200 visitors (notice it's slightly bigger)
- 👤👤👤 = 300 visitors

3. Combined Approach

- 📦 = 100 units sold
- 📦📦 + 50 = 250 units sold

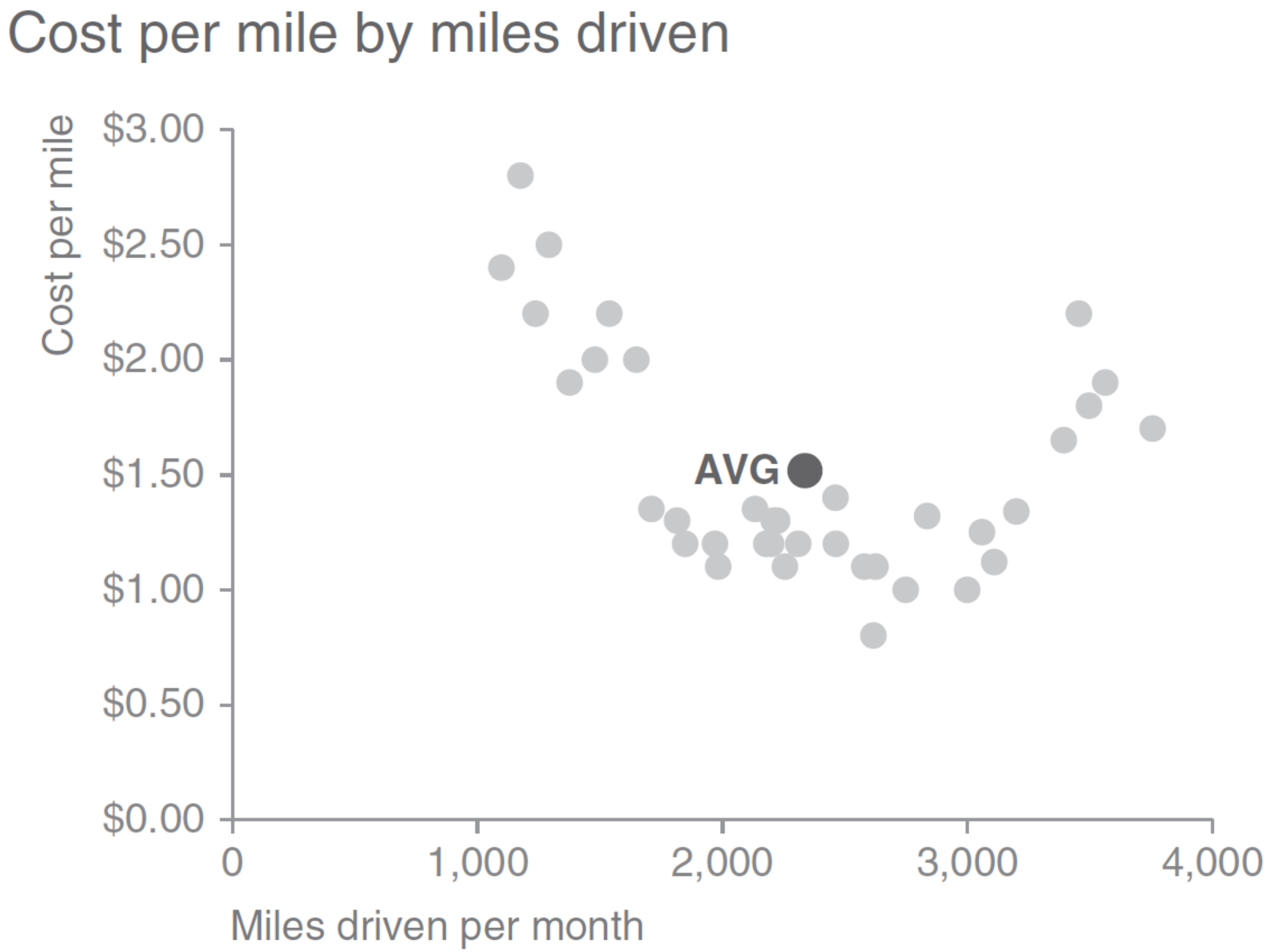
Scatter plot

Used to observe and visually represent the relationship between two numerical variables

Patterns:
Scatter plots help reveal patterns or relationships between the two variables. Positive, Negative, or No Correlation.

Strength of Relationship:
The closer the dots are to forming a line or curve, the stronger the correlation. Widely scattered dots indicate a weaker correlation.

Outliers:
Points that lie far away from the other data points can be easily identified as outliers.



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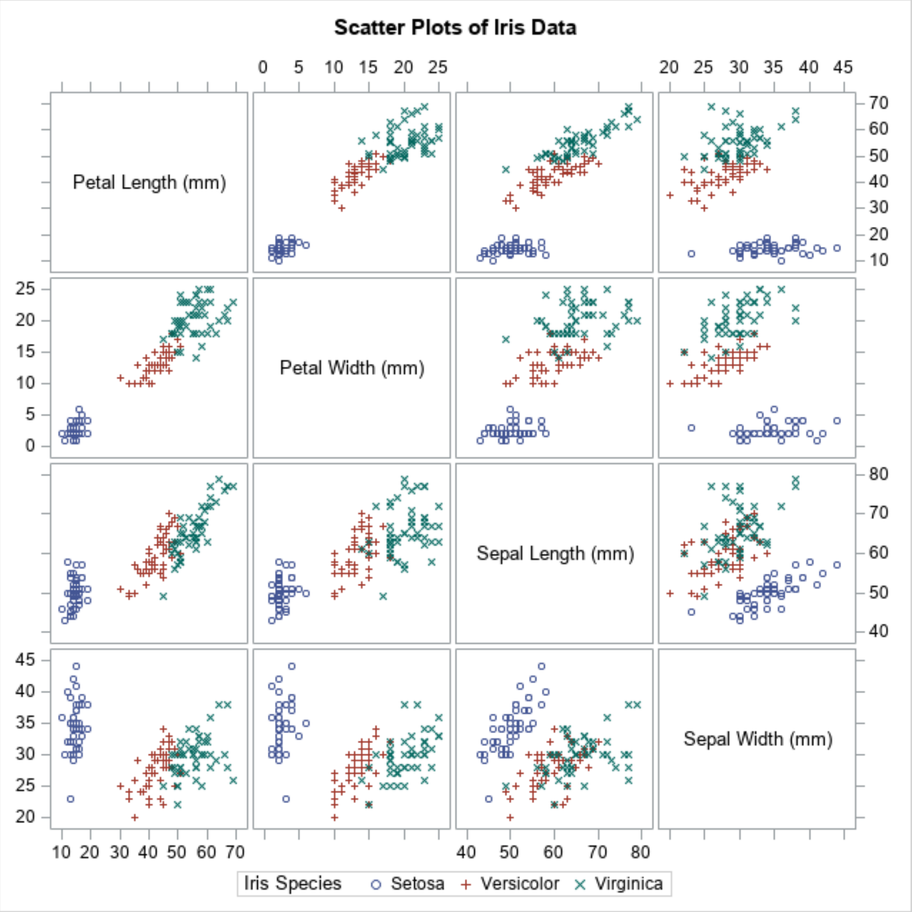
Scatter Plot Matrix (SPLOM)

A grid (or matrix) of scatter plots

Overview of Pairwise Relationships: Provides a compact view of relationships between all pairs of variables.

Pattern Identification: Helps in identifying correlations, clusters, and potential outliers.

Understanding Data Structure: Gives insights into the underlying structure of the data.



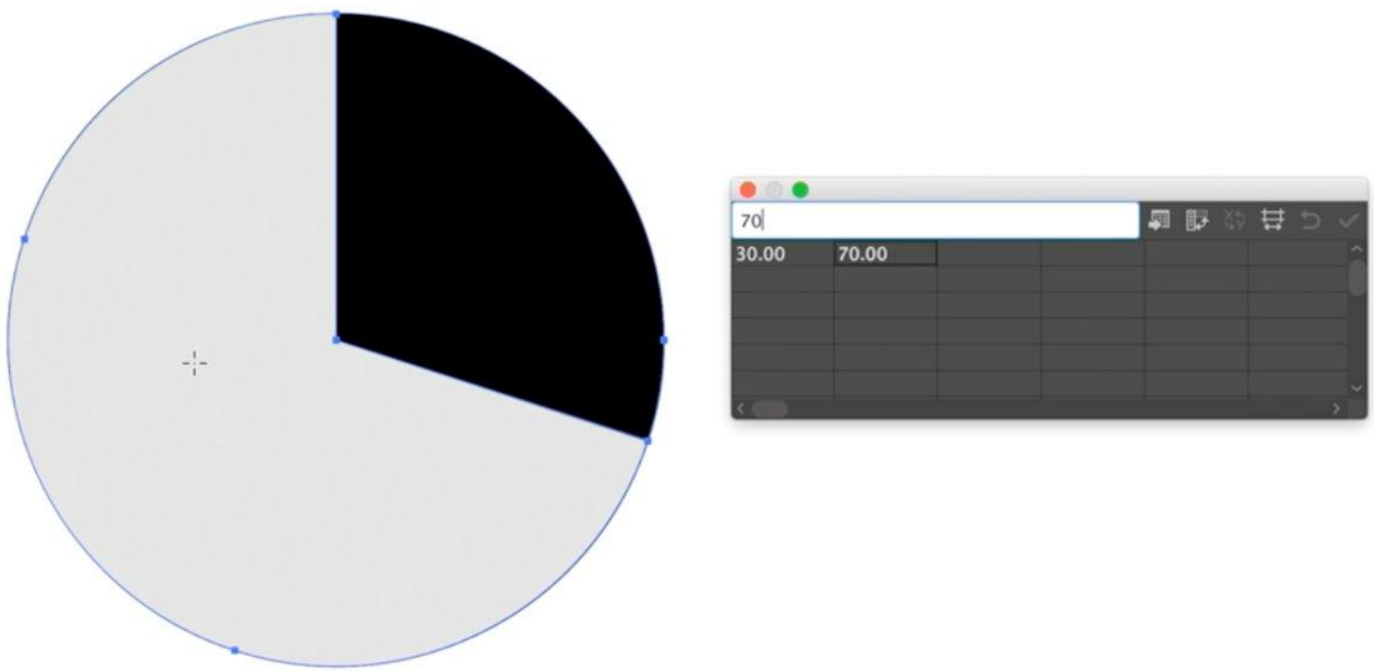
Source: https://blogs.sas.com/content/iml/2019/04/01/matrix-operations-by-groups.html

Pie charts

Strengths of Pie Charts

Quickly convey proportions: Pie charts make it easy to see the largest and smallest portions at a glance.

Simple and visually appealing: Their circular shape is often more engaging and visually appealing for audiences.



Polar area diagram

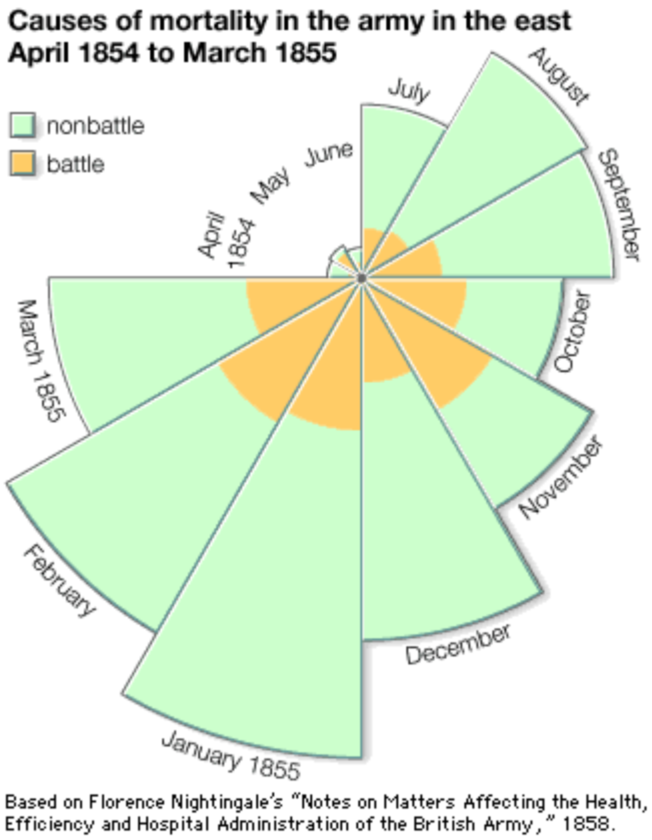
Similar to pie chart except sectors are equal angles and differ in extension from central point

Nightingale's Coxcomb chart

Central Point: The center of the circle is the origin from which all segments radiate

Segments: Each colored segment represents a month

Area of Segments: The area of each segment is proportional to the number of deaths that occurred in that month. Larger area means more deaths. Crucially, it's the area, not the length of the segment, that matters.



Source: https://kids.britannica.com/students/assembly/view/70822

Image from: https://www.americamagazine.org/politics-society/2017/09/21/what-john-courtney-murray-would-say-about-state-public-debate

Section 4

Insights to Drive Action

Introduction to insights

What an insights is?

Insights are meaningful interpretations drawn from data — they reveal patterns, relationships, or shifts that aren't obvious at first glance

Why insights matter?

They turn information into understanding, helping us prioritise what's important and make more confident, informed decisions

How insights emerge?

Through analysis and visualisation, we uncover deeper meaning — visuals help highlight trends, comparisons, and anomalies that lead to insight.

Top Tip: The "So What" Test

How we turn data into insight

The Building Blocks: The raw ingredients and steps that turn data into insight.

The Thinking: How we analyse, compare, and interpret patterns.

The Output: What an insight looks like when it's communicated clearly.

Types of insights in data analysis

Descriptive:
What happened? (e.g. summarising past events)

Diagnostic:
Why did it happen? (e.g. identifying causes or factors)

Predictive:
What might happen next? (e.g. forecasting future trends)

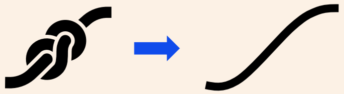
Prescriptive:
What should be done? (e.g. recommended actions)

Activity Time: Guess the Profession

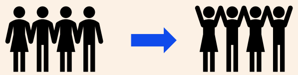
How visuals strengthen your insights

Visuals help audiences understand, trust, and act on your insights.

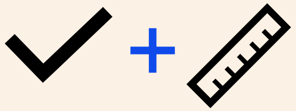
Clarity:
Makes complex information easier to understand at a glance.



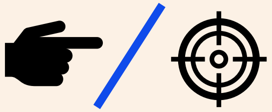
Engagement:
Captures attention and helps insights stick.



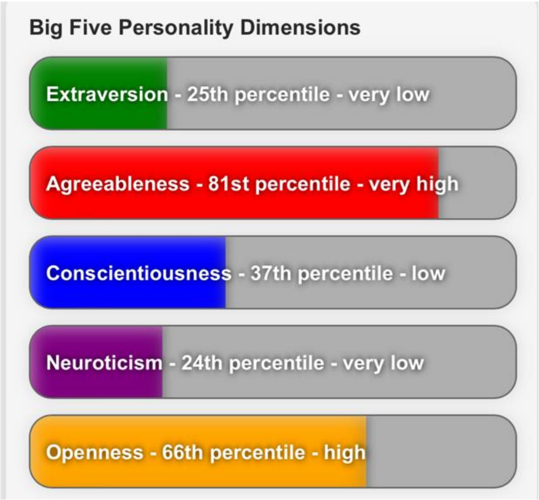
Accuracy:
Represents data truthfully and avoids misleading interpretations.

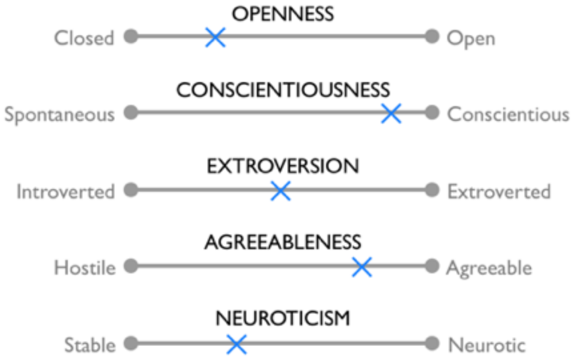


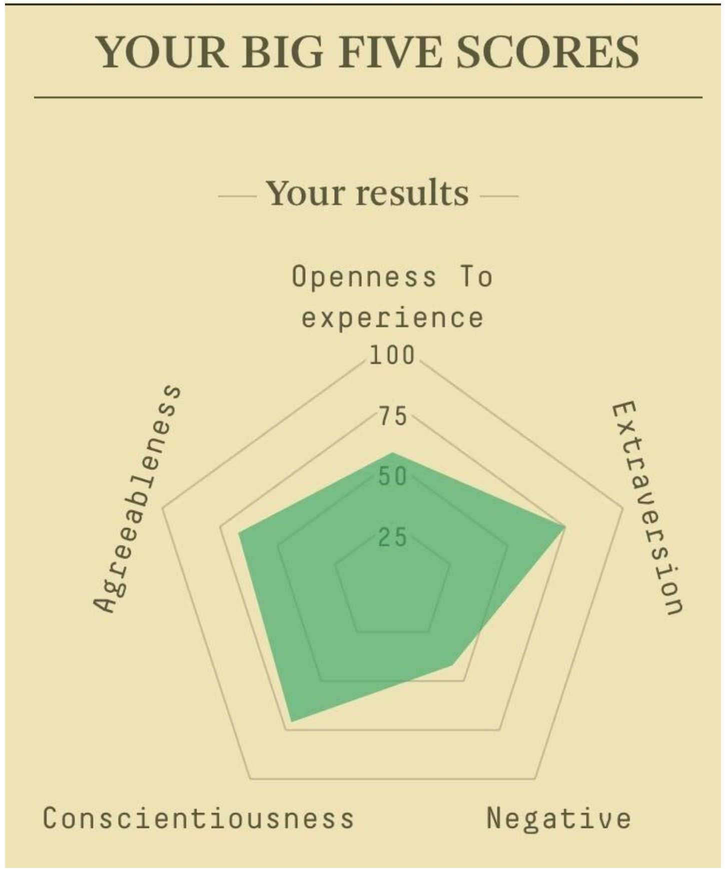
Focus:
Directs the audience to the key message without distraction.

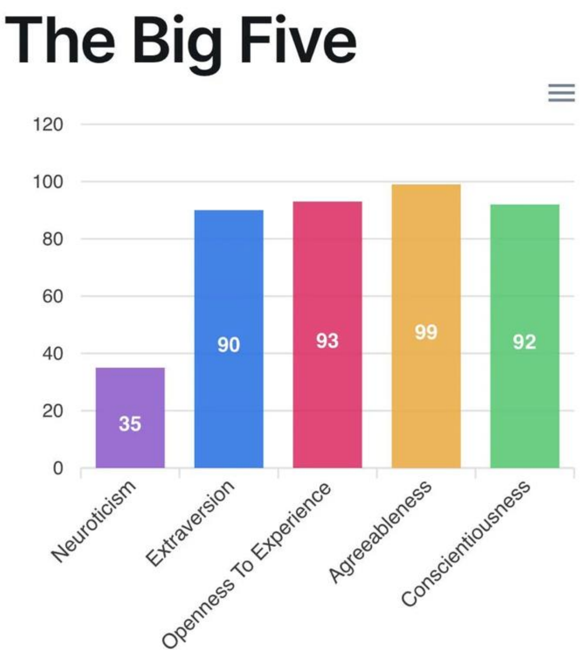


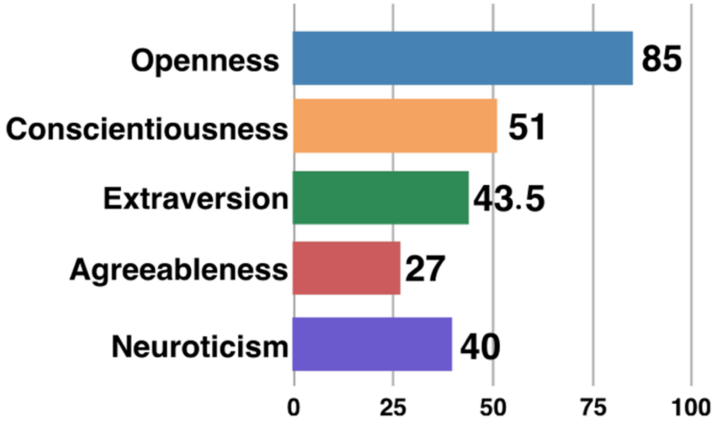
Which visualisation do you prefer?



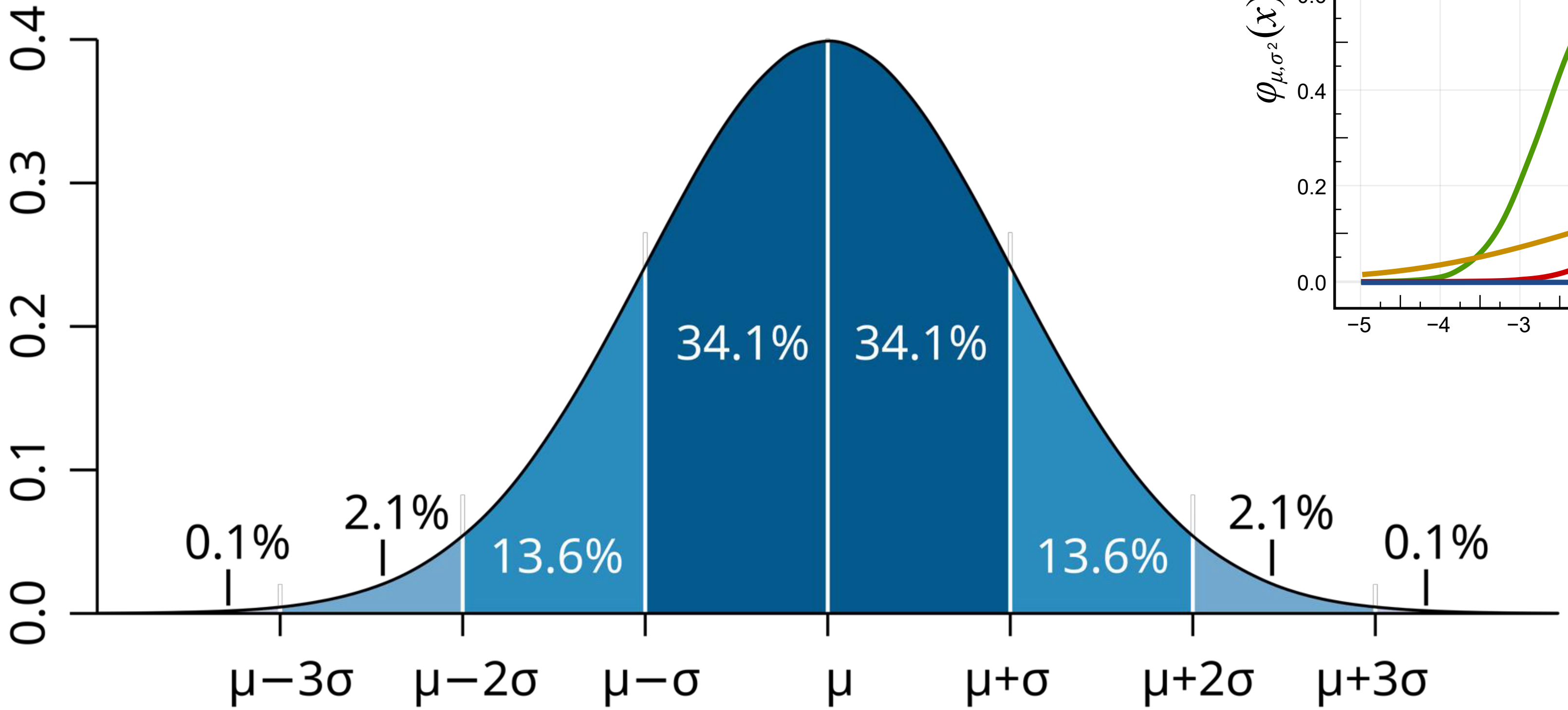


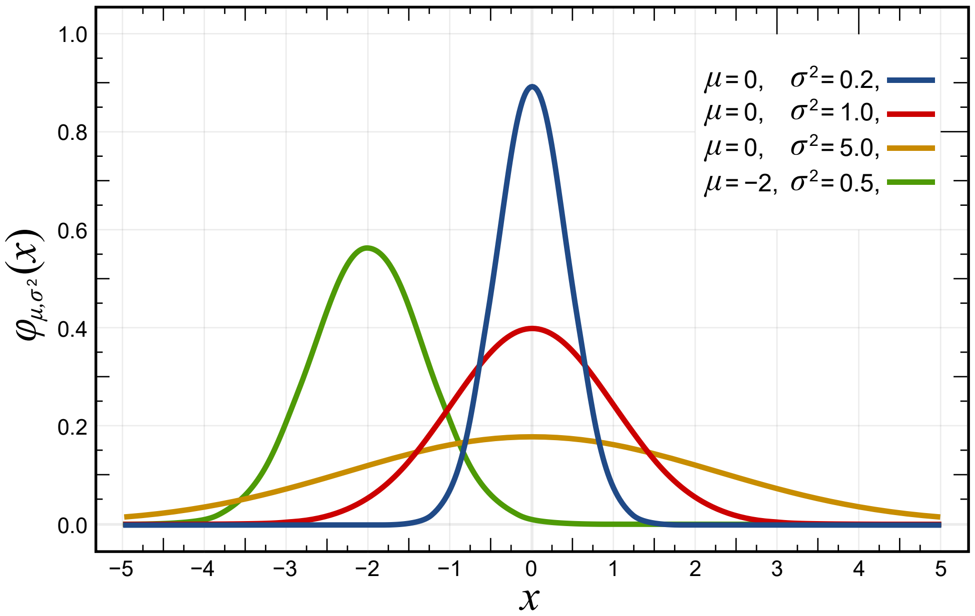






Key Concept: Normalisation

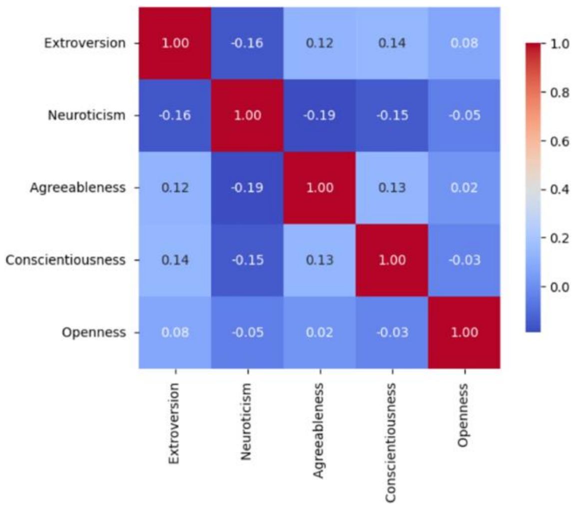


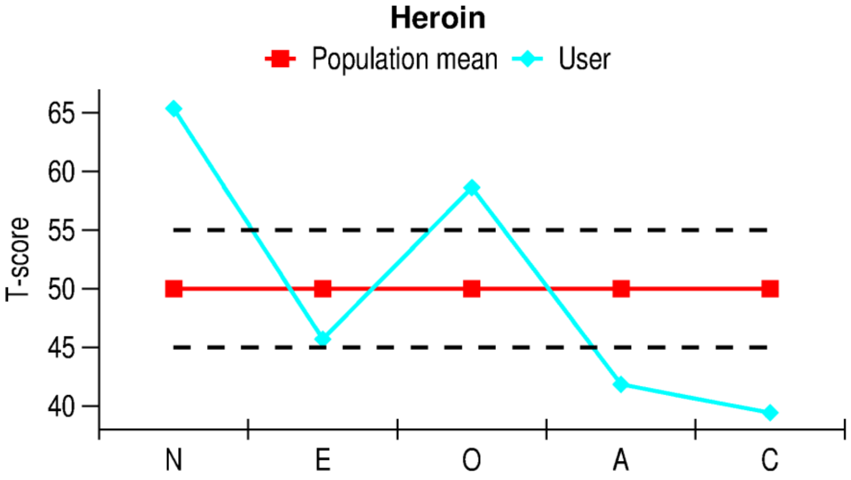


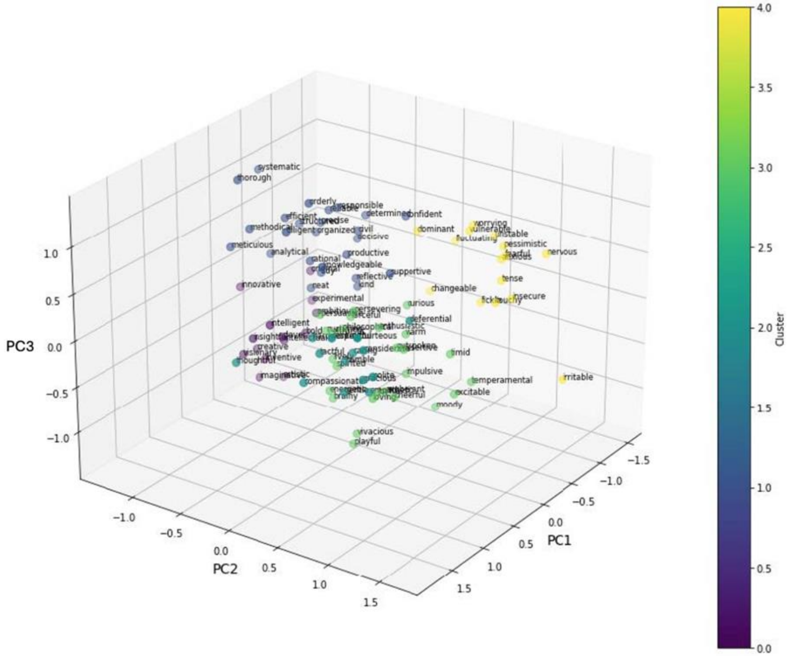
$$Z = \frac{X - \mu}{\sigma}$$

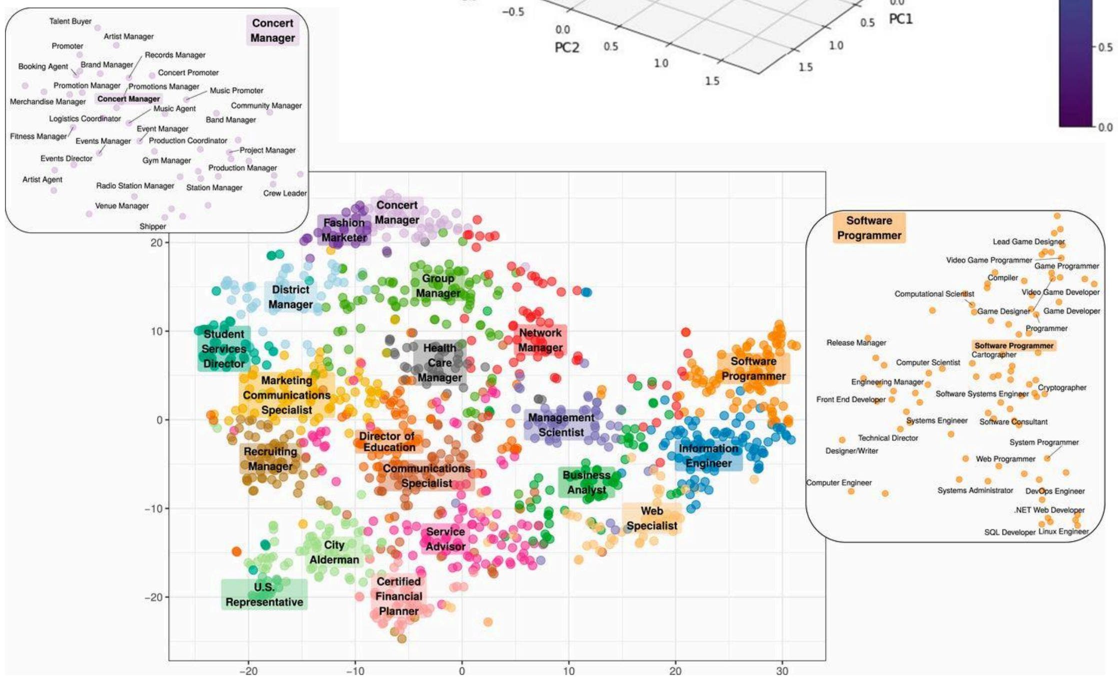
$$t = \frac{X - \mu}{s/\sqrt{n}}$$

What about these?



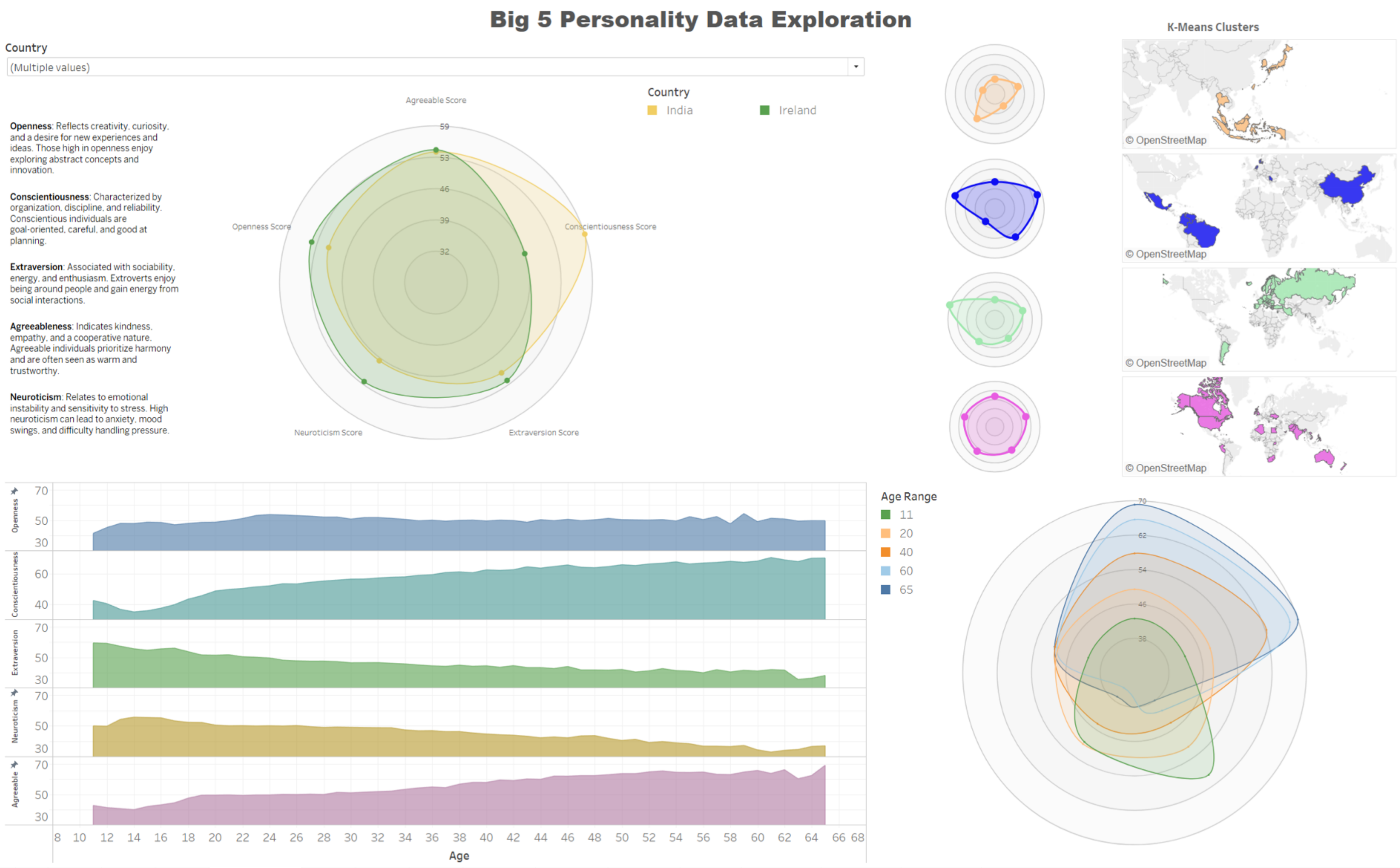




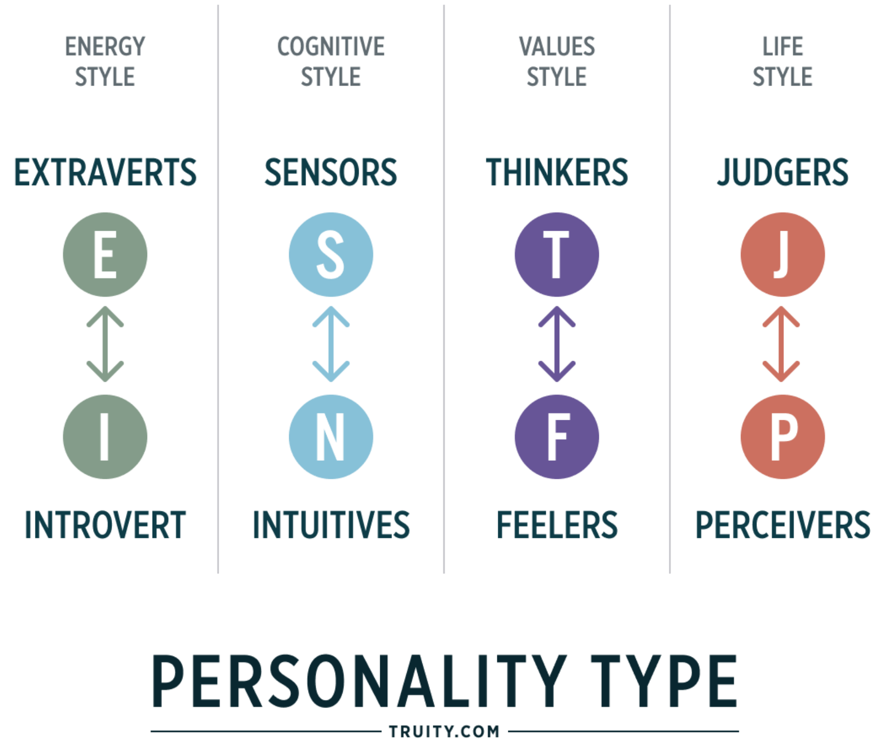


What about Tableau?

r/dataisbeautiful

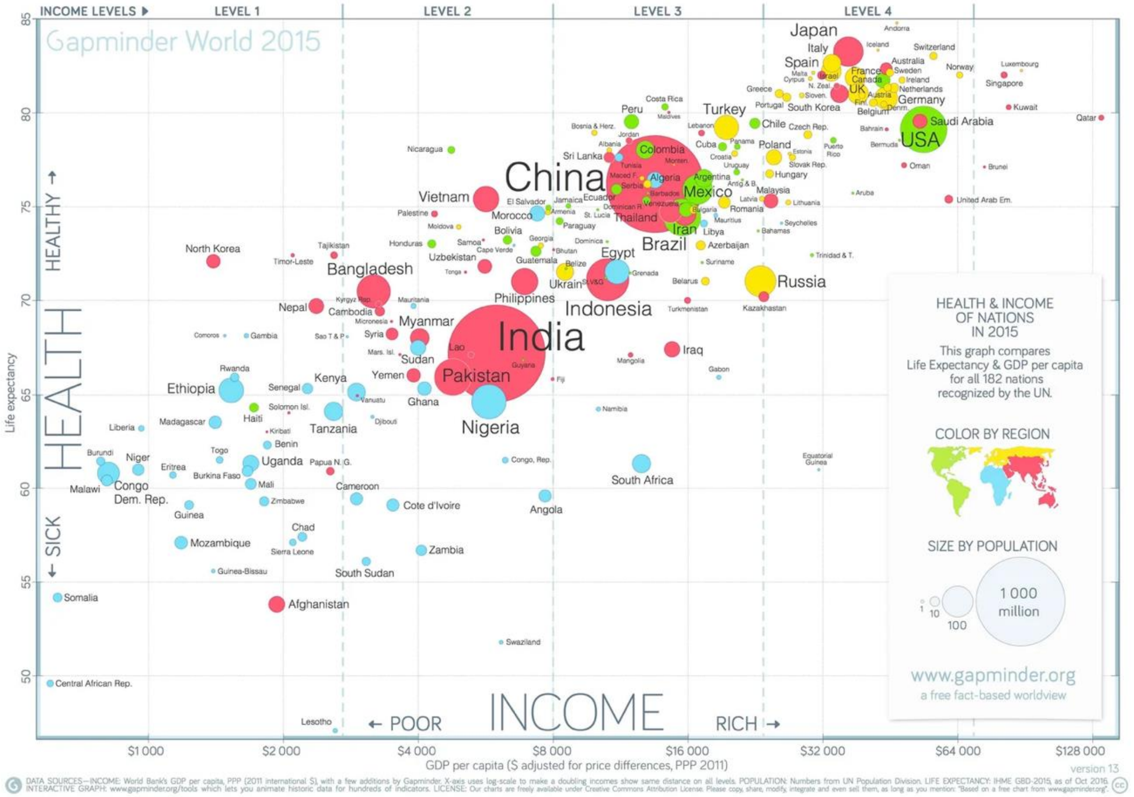


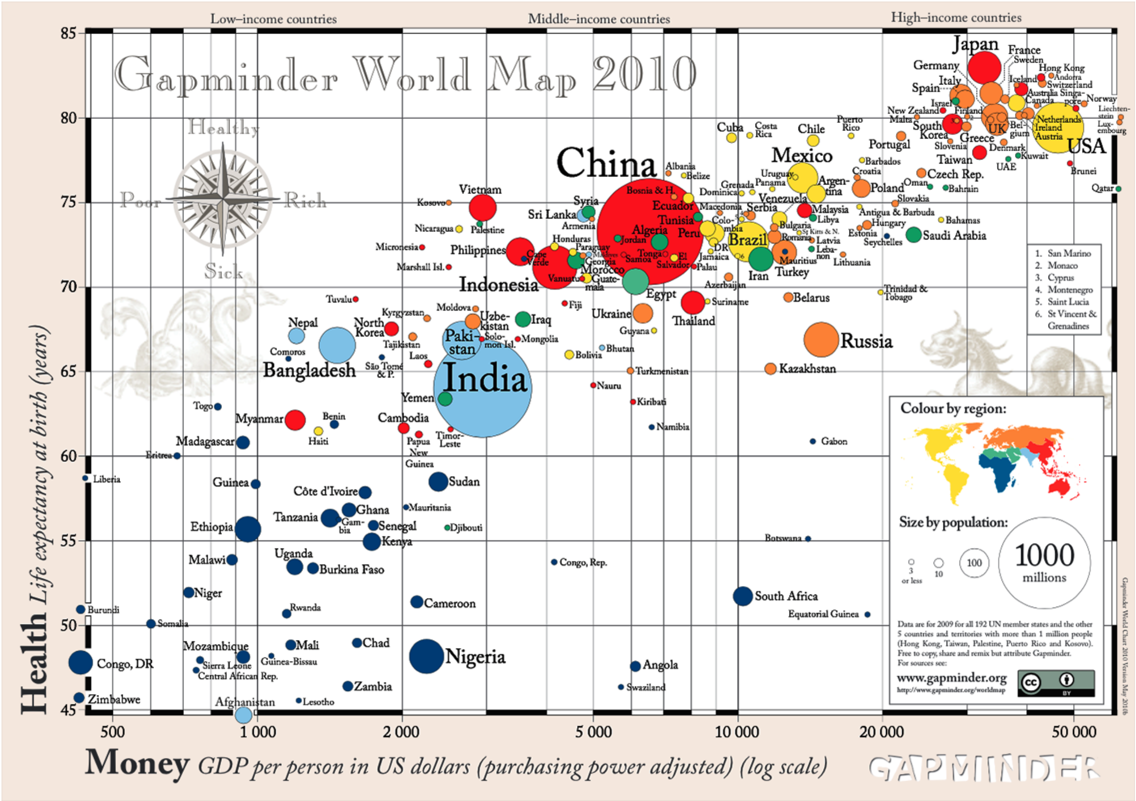
Discrete vs Continuous Data



How would you visualise MBTI results differently from OCEAN results?

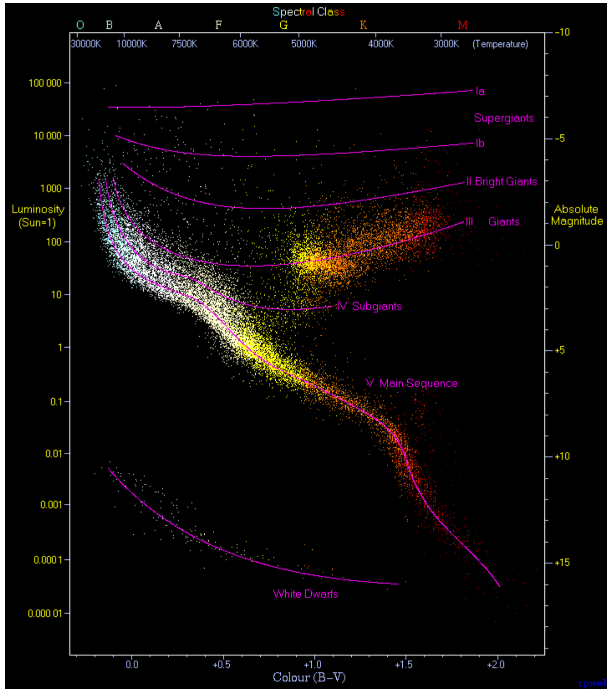
Discrete vs Continuous Data

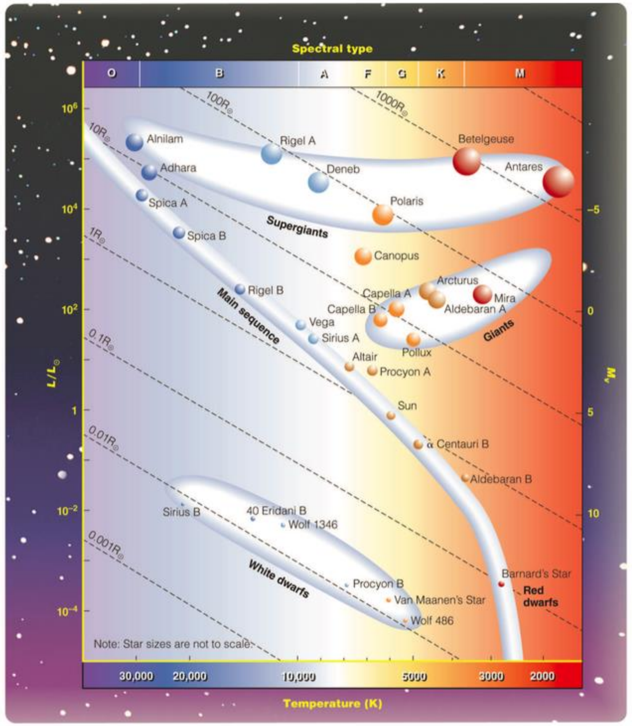


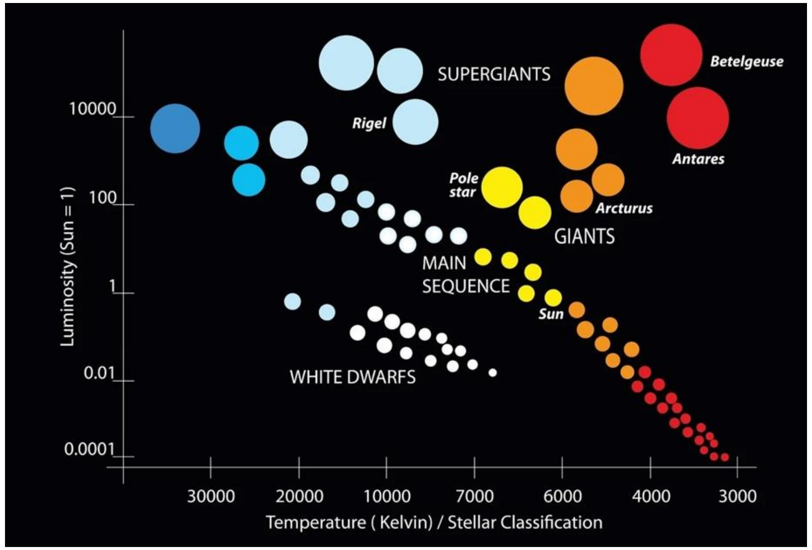


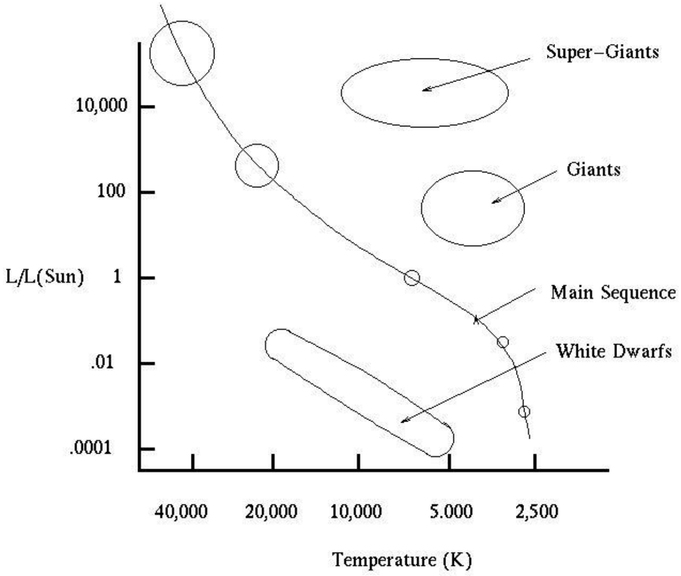
How many dimensions are being visualised?

Discrete vs Continuous Data









How many dimensions are being visualised?

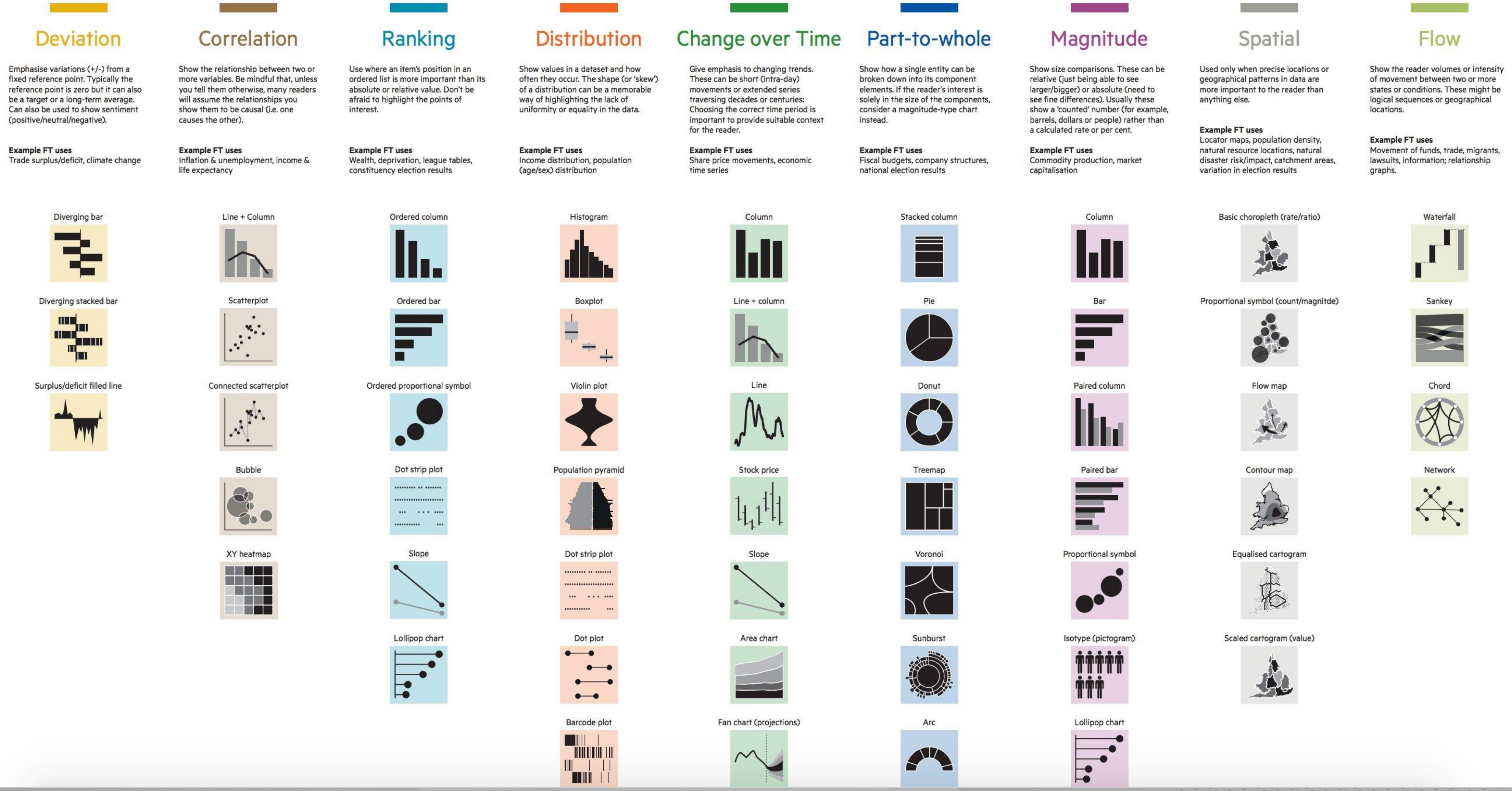
See also http://astro.wsu.edu/worthey/astro/html/lec-hr.html

Data Types

Data Type	Computer Data Type	Python Implementation	Example / Real-world Use
Nominal	***string, char, bool***	`object / string`	*"Red", "Blue", "Green"*
Ordinal	***int (mapped)***	`pandas.Categorical`	*"Bronze", "Silver", "Gold"*
Count	***int, uint32, long***	`int`	*number of items in a cart (e.g., 5)*
Continuous	***float, double***	`float64`	*person's height (e.g., 175.5 cm)*
Time / Date	***datetime, timestamp***	`datetime64[ns]`	*2026-02-24 08:30:00*
Duration	***timedelta, interval***	`Timedelta`	*45 minutes, 12 seconds*
Text	***string, clob***	`object / string`	*blog post*
Audio / Video	***blob (Binary Large Object)***	`bytearray / bytes`	*recording.mp4*
Image	***matrix, array***	`numpy.ndarray`	*pixel grid*
Multimodal	***Vector / Embedding***	`torch.Tensor / ndarray`	*video with AI-generated caption*
Spatial / Geometric	***point, polygon, mesh***	`geopandas / shapely`	*GPS coordinates or 3D model*

Visual Vocabulary

https://gramener.github.io/visual-vocabulary-vega/



Recognise these tools?

UTS

Thank you!